Designing for Emergence:

A Human-Centered Approach to Emergent Strategies in Interaction Design through Video Games

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5/30/25

Master's Thesis
Digital Design

Abstract

This thesis addresses the underexplored role of emergent phenomena in design by investigating how emergence can provide methodological value in interaction design through the development of a structured framework informed by ethnographic fieldwork and theoretical analysis. It reframes emergent design as a deliberate and human-centered methodology rather than a secondary byproduct of iterative processes by drawing on meta-design principles in order to treat users as co-creators. Competitive fighting games and their community serves as a case study to examine how unpredictable user interactions give rise to appropriated emergent strategies and behaviors that can inform design innovation. Through ethnographic observation and participant interviews within the Fighting Game Community in Aarhus, the study studies how complex systems such as competitive games foster a rich interplay between rule structures and user emergent behavior. Players routinely exploit, adapt, or reconfigure mechanics in ways not foreseen by the original developers, demonstrating the creative capacity of end-users to expand systems innovatively. These user-driven phenomena inform the development of a design framework, created to guide designers in cultivating systems that allow for emergent outcomes. By embracing user-driven emergence, the proposed framework enables designers to intentionally harness emergent user innovations.

The thesis introduces the concept of situated conditions as a way to describe the interplay between context, identity, and temporality in shaping how emergence is shaped, which was discovered during the ethnographic field research. Rather than viewing emergent outcomes as only the result of system affordances, the concept of situated conditions emphasizes how social contexts, personal identity, and time all contribute to the development of new strategies and uses among system users. These conditions are not fixed but continuously produced through use, influencing how user innovation takes place.

Furthermore, the study emphasizes that emergent design requires a shift in the designer's role: as facilitators of co-creation. Rather than controlling outcomes, the designer sets the stage for emergent interactions by leaving space for appropriation, enabling consequence-

laden decision-making, and fostering environments where user contributions are made visible. This approach supports the creation of interactive systems that evolve alongside their users, effectively bridging emergent phenomena and human-centered design practice.

Ultimately, this thesis contributes both a theoretical foundation and practical framework based on ethnographic findings, in order to argue for emergent design as a distinct methodology within the field of interaction design.

Contents

1. Introduction to Emergent Design

Emergent design continues to be a largely underexplored idea within the realm of interaction design research. Although the concept of emergence is broadly acknowledged, the specific use of emergent design as a methodological framework is rarely formalized. Instead, people often view it as a secondary component of various design processes, not as a distinct methodology. This research gap prompts the main research question that inspired this paper: *how can emergence provide value to a design process as a methodological framework in interaction design?*

One example of an attempt at applying emergence, is seen in the work of Defazio and Rand (2011), who discuss emergent design within the context of serious games for healthcare education. They describe emergence as a problem-solving mechanism that evolves through interaction and feedback, allowing systems to adapt over time (Defazio & Rand, 2011). However, their research primarily attributes emergence as a biproduct of iterative development rather than emergence as a source of iterative solutions. While iteration is desirable in many traditional design methods, emergence may offer an alternative lens to enhance and expand iterative processes. This distinction shows the need to investigate emergent design as a structured methodology rather than a mere biproduct from already existing approaches. Given that video games provide an interactive medium where players engage with complex systems, they serve as curious environment to study emergence in design. Video games naturally facilitate emergent behaviors, as players interact with mechanics in ways that often go beyond the designers' original intentions. An example of emergence shaping game design can be found in Street Fighter II by CAPCOM (see figure 1). During development, a bug allowed players to cancel animations early, leading to the accidental discovery of combos as a game mechanic. What could have been dismissed as an unintended flaw instead became a defining feature of the fighting game genre. Then, how can emergent behaviors, when recognized and leveraged by designers, significantly impact and improve interactive systems and user experiences?

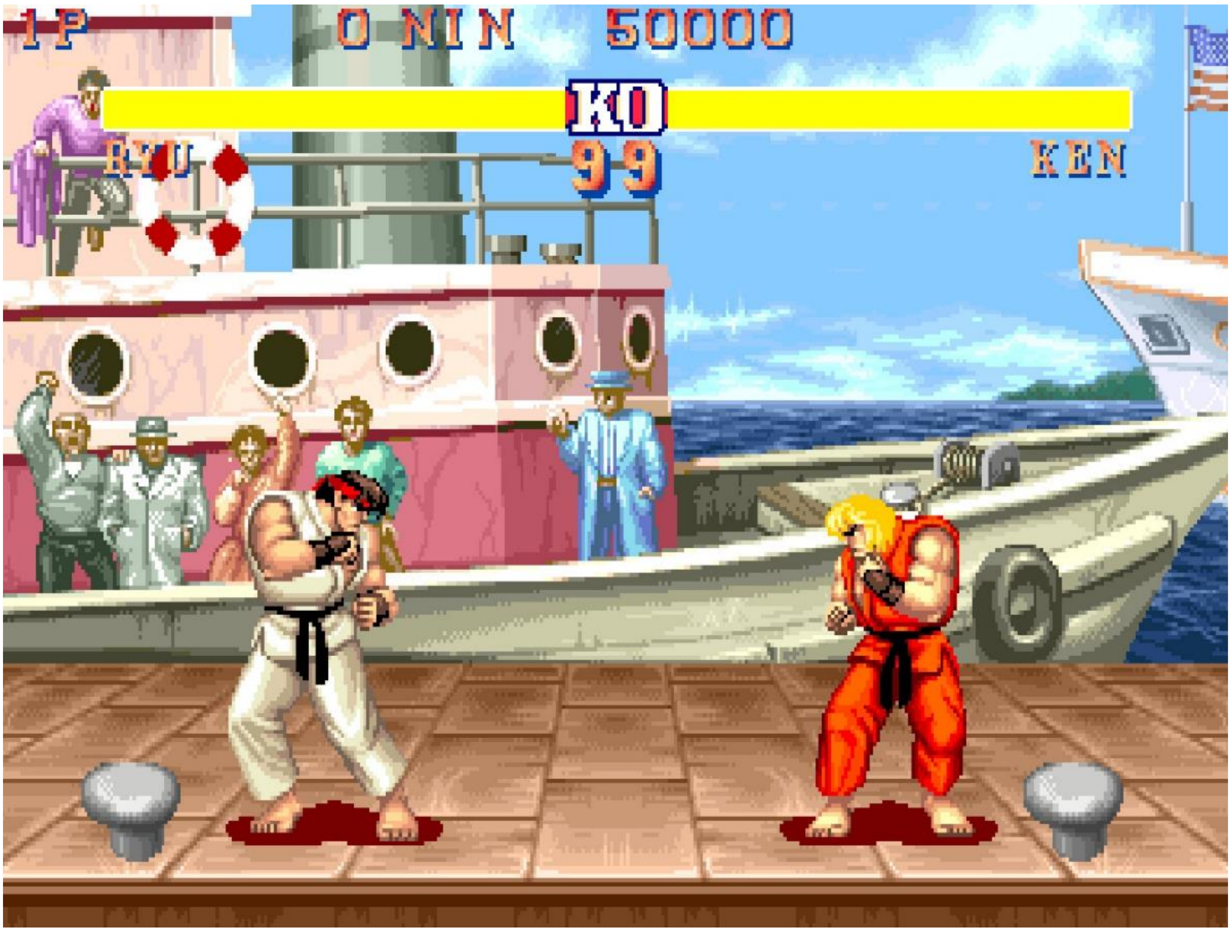

Figure 1 - Street Fighter II by CAPCOM

Existing work touches on emergence as a byproduct of iterative design, but no framework seemingly treats emergence as a guiding design principle, which is a gap this paper aims to fill. In doing so, this research aims to explore emergent design as a methodological framework, examining its impact on interaction design through video games. The objective is to develop a structured approach that leverages emergence to enhance user experience and foster innovation. To achieve this, the paper will propose a baseline framework grounded in human-centered design values, interaction design, and game design literature. This framework will be further refined using data gathered through ethnographic field research, including interviews and empirical field studies, to present an emergent design framework and methodology that can be applied to various interactive systems. By doing so, this study seeks to bridge the gap between emergence as a phenomenon and emergent design as a structured methodology, ultimately broadening the scope of interaction design. In the following sections, this paper reviews the current understanding of emergence as well as foundational concepts in interaction design and game studies, then examines emergence in games before proposing the first framework for emergent design. Following this, it goes over the field research performed and analyzes its data to ultimately propose an iterated version of the emergent design framework, of which it briefly discusses its relevancy in current design environments.

2. Exploring Emergence

The observation of emergent behavior in video games, specifically those in the fighting game genre, is what prompted the interest in exploring its role in design processes. Therefore, the general understanding of emergence and its relevancy should be explored.

In his work "The Emergence of Everything," Harold J. Morowitz (2002) describes emergence in broader terms as a phenomenon where unforeseen properties develop from the interactions of simpler elements (p. 20). Morowitz states that "the whole is somehow different from the sum of the parts" (Morowitz, 2002, p. 20), suggesting that emergent properties cannot be solely anticipated by comprehending the individual components. This idea is compared to biological evolution, where "novelty occurs by mutation, translocation, selection, and differential survival" (Morowitz, 2002, p. 20), indicating that emergence is a crucial aspect of both natural and human-made systems. Additionally, he emphasizes the importance of emergence by asserting that it "develops previously unrealized ways of deepening our understanding of the past eons and illuminates how the universe, after a lengthy and intricate 12-billion-year journey since the Big Bang, has led to the emergence of the human mind and modern humans" (Morowitz, 2002, p. 16). Furthermore, Jeffrey Goldstein (1999) defines emergence as the occurrence of new and coherent structures arising spontaneously from the interactions of simpler components within a system. Unlike conventional top-down methods, emergent phenomena cannot be reliably predicted just by examining individual components in isolation, highlighting interaction and self-organization as fundamental aspects. Essentially, emergence embodies the unforeseen; the results we prepare for but that unfold in unexpected ways, or the challenges that present themselves in the moment and necessitate our immediate adaptation. In the realm of digital games, Juul (2005) highlights that although unforeseen events can occur that the designer may not have anticipated, these incidents remain innately connected to the game itself:

> “When playing a game, events may occur that the designer did not predict, but even if many emergent and non-intentional events can happen in a game, they still would not happen without the game – it is the game that allows them to happen” (Jesper Juul, 2005, p. 200)

This highlights the unique capacity of video games, where emergent behavior is essential to the experience. Unlike other forms of media, video games provide an interactive space where players' direct interactions can create unforeseen outcomes, thus enabling a unique level of complexity and creativity compared to other traditional narrative media.

2.1. Current Emergent Approaches in Interaction Design

While emergent design remains an underexplored concept in interaction design research, recent studies have begun to explicitly investigate and experiment with emergence as a fundamental design principle. A notable example of such research is provided by Murray-Browne and Tigas (2021), who introduce the concept of *emergent interfaces* (p. 15). Informed by complex systems theory, emergent interfaces are designed using unsupervised machine learning techniques, allowing system behaviors to dynamically evolve from real-time user interactions rather than adhering to fixed interaction paths predetermined by designers. Emergent interfaces leverage contextually derived emergent representations, (Goldstein, 1999) meaning their responsiveness and user interactions continually adapt to changes in context, therefore exemplifying how emergent principles can drive adaptive, personalized experiences (Murray-Browne & Tigas, 2021). As Bennett et al. (2021) indicate in their article on emergent interactions, complexity-oriented approaches in human-computer interaction (HCI) that highlight emergent behaviors arising from interactions are gradually gaining recognition, though they remain peripheral to traditional design methodologies.

Further reinforcing this trend towards emergence is recent research by Gaver et al. (2022), who explore emergence specifically within the methodological framework of practice-based design research. Gaver et al. (2022) also recognize that emergence has long been acknowledged as a central, yet under explored feature of practice-based design research. They argue that emergence is in fact a generative condition of inquiry within design contexts. Furthermore, Gaver et al. (2022) generally question the prevalent narrative frameworks in HCI that prioritize linearity and predictability too much. These traditions run the danger of distorting the dynamics of design work, which often requires ongoing reinterpretations and re-framings. From their experiences, the quality and significance of the results and insights of effective design study characterize it, rather than its conformity to pre-defined objectives, no matter which route led it there (Gaver et al., 2022). They support this by drawing on Lucy Suchmann's (1987) interpretation of ethnomethodology to explain emergence as arising from interactions between actors, skills, environments, and materials, rather than following these more traditional linear plans. Suchman's (1987) concept of *situated actions* is used to argue that plans are not prescriptive but orienting, as they guide actions that are often reformulated during practice. This creates room for assessing design projects depending on their situated showings and their outcomes within the changing surroundings.

Furthermore, according to Gaver et al. (2022), design processes themselves can be emergent: "Methods, tactics, goals, and even topics can unfold and change as researchers adapt and learn in the course of their projects" (Gaver et al., 2022, p. 1). This highlight just how such emergent approaches contrast with the systematic methodologies typically employed in design research, often creating tension between the unpredictable nature of emergent outcomes and conventional expectations for structured design processes. This paper chooses to use this term *emergent outcomes*, as an umbrella term for all system outputs originating from unexpected behavior. This can be emergent strategies, inputs and outputs, styles, norms, etc. produced by user interactions.

Gaver et al. (2022) also propose embracing an open-ended, iterative approach to research, suggesting that by acknowledging and working with emergence, designers can uncover new insights, directions, and possibilities that were not anticipated initially. This methodological perspective together with Murray-Browne and Tigas's (2021) work, suggests that emergent design extends beyond facilitating dynamic user interactions. It also fundamentally shapes how designers themselves approach and adapt to complexity within the design process. By explicitly acknowledging emergence as both an outcome and a process, this research shows the potential for an emergent design approach to produce more user-friendly and innovative outcomes compared to traditional, top-down design practices, further establishing emergent design as a promising research direction within interaction design.

3. Emergence from the Perspective of Interaction Design

At its core, *Interaction Design* is about interactions between a user and a system. Helen Sharp, Jenny Preece, and Yvonne Rogers (2019) describe interaction design in *Interaction Design: Beyond Human-Computer Interaction*, as the process of designing interactive products and systems that aim to support and enhance the way people interact with technology (p. 9). The defining qualities of interaction design are grounded in the principle of designing interactions (inputs and outputs) within interfaces that are intuitive and efficient. By understanding users' behaviors, interaction design seeks to ensure that technology serves human needs effectively (pp. 13-16). These qualities include a strong emphasis on usability, ensuring that systems are easy to use and allow users to complete their tasks with minimal effort and confusion (Sharp et al., 2019, p. 19). Furthermore, Sharp et al. describe interaction design as “multidisciplinary, involving many inputs from wide-ranging disciplines and fields” (Sharp et al., 2019, p. 32), highlighting its versatility as a broad design approach that can be influenced by various contexts, including video games. At its core, video games are defined by their higher level of intractability compared to other media. Interaction design and video games therefore share a core element, which is a focus on inputs and outputs.

3.1. Appropriation in Interaction Design

In HCI and interaction design, appropriation refers to how users adapt or reconfigure technologies beyond their designers’ original intentions. A concept seemingly similar to emergence; showcasing the relevancy of emergence within the field of interaction design. Alan Dix (2007) characterizes appropriation as users employing systems in “ways never envisaged by the designers,” (Dix, 2007, p. 27) emphasizing that no design can anticipate every possible use. Rather than being simple misuse, this creative repurposing is often viewed as a positive phenomenon, revealing how technology can serve unanticipated needs and contexts. Antti Salovaara (2008) similarly defines appropriation as the process of inventing new purposes of use for technologies through new user understandings of the affordances within digital artifacts, noting that it is a pervasive aspect of how people interact with tools in everyday life. For example, using a digital camera as a makeshift mirror is an illustration of a user appropriating affordances of an artifact for an unintended function (Salovaara, 2008).

This kind of repurposing shows users' creativity and is considered a sign of good design when a product makes new uses easy to invent. Salovaara (2008) also argues that appropriation is often community-driven, and visibility plays a key role in adoption and innovation. One might spot another passenger on the same train as themselves using their repurposing the front-facing camera of their smartphone as a mirror, then be inspired to do it themselves later. By observing appropriation in practice, designers can gain insight into new solutions. End users can discover improvements or novel applications that designers had not, or could not, imagine, which aligns with the unanticipated outcomes of emergence, showing how it can enrich an interactive system.

Designers can support and encourage appropriation through flexible design. Dix (2007) argues that although one cannot design the unexpected, one can "design to allow the unexpected" (Dix, 2007, p. 27) by building systems that invite user interpretation and modification. In their guidelines for designing for appropriation, Dix (2007) outlines several strategies that make interactive products more amenable to user adaptation and creative use:

- Allow interpretation: Avoid fixing rigid meanings to every feature; provide elements that users can assign personal meaning to.
- Expose internals: Make system processes and feedback visible so that users understand how the technology works and can appropriate those workings.
- Support, do not control: Design for flexibility rather than prescribing one correct use.
- Enable reconfiguration: Ensure the system is tailorable or modular so that users can reassemble components or extend functionality.
- Encourage sharing: Provide mechanisms for users to share their novel uses or tips with others.

By incorporating these principles, designers can invite users to participate in the ongoing evolution of the system. Appropriation therefore becomes a bridge between user behavior and design innovation, as users' unanticipated practices inform future design improvements. These qualities are in line with what this paper tries to argue for emergence. Therefore, in the context of emergent design, appropriation could be a key mechanism by which new functionalities or patterns arise from use: people co-create new interactions that were not explicitly designed upfront. This perspective reinforces that design can accommodate and learn from the inventive ways users make technology their own.

3.2. Meta-Design as a Foundation for Emergent Design

While appropriation highlights how users adapt designs after the fact, *meta-design* (Fischer & Giaccardi, 2006) offers a proactive approach to cultivate adaptations from the start. Meta-design is a framework introduced by Fischer and Giaccardi (2006). It focuses on creating open systems that end users can modify and evolve over time. Essentially, a meta-designed system is not a finished artifact but a platform for ongoing design as it is built with the assumption that users will act as co-designers. This concept has its origins in *participatory design* (Blomberg et al., 2017), as it wants to involve users in the initial design process as co-creators. Meta-design builds on this by extending design beyond the development phase into the actual use of the system. Meta-design keeps the design space open, enabling a co-adaptive process between users and the system over its lifetime. Fischer and Giaccardi (2006) argue that future uses and problems cannot be completely anticipated while designing, so designers must intentionally leave room for users to adapt the system to their situational needs. In practical terms, this means embracing *underdesign* (Fischer & Giaccardi, 2006, p. 7): creating an initial structure that is deliberately incomplete or flexible, so that users can help finish the design by tailoring the system to local contexts and emergent requirements (Fischer & Giaccardi, 2006, p. 7). A key principle of meta-design is that designers must be willing to share control with users regarding how the system evolves. Instead of overdesigning, designers should expect that users will surprise them, and they welcome those surprises as part of the system's ongoing growth. This approach aligns with Fischer et al.'s (2004) *Seeding, Evolutionary Growth, Reseeding model,* where an initial design seed is released, then the community of users evolves it through use and iterative improvements, and occasionally designers intervene to reseed or reorganize the accumulated changes (see figure 2).

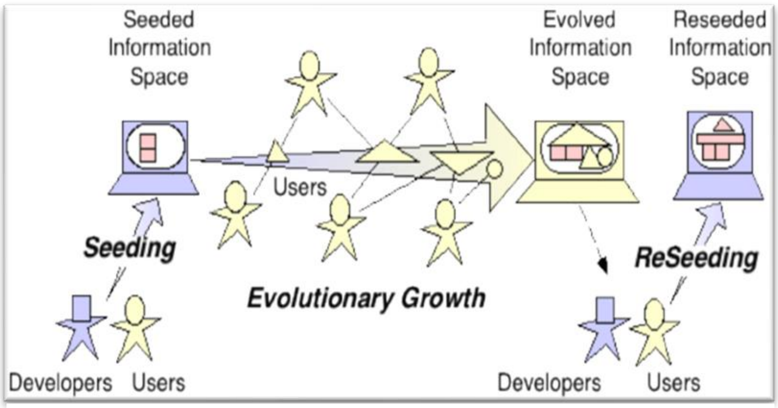

Figure 2 - SER Model (Fischer et al., 2004)

The success of open-source software provides real-world validation for meta-design: given the right open infrastructures, large groups of users can collaboratively create new features, content, and improvements that no single design team could have fully envisioned in advance. In HCI literature, this is described as a shift from a passive consumer culture to a culture of participation, where users expect to actively shape their tools and environments (Fischer, 2011). By designing interactive systems as open, evolvable platforms, meta-design intrinsically supports emergent behaviors and solutions arising from its community of users.

3.3. Human-Centered Design

While not explicitly under the umbrella that is interaction design, Human-Centered Design (HCD) will provide a basis from which the humanistic perspective the researcher chooses to approach the future research from. Due to the user-centric context of the future empirical research and interaction design itself, it is important to understand the values associated with Human-Centered Design as an approach as they are aligned.

As discussed by Fancesca Tosi (2020), Human-Centered Design (HCD) is a design philosophy that prioritizes understanding and addressing the needs and experiences of users throughout the design process (p. 48). It is characterized by an iterative, research-based approach that begins and ends with the user, ensuring that the final product effectively meets their requirements (Tosi, 2020, p. 48). In their book, Tosi (2020) describes what they refer to as six fundamental principles for the HCD approach to design:

1. “the design is based on an explicit understanding of users, activities and environments” (Tosi, 2020, p. 50)

Reading into this shows the primary aim of HCD, which is to create solutions that are tailored to the specific needs of users, ultimately enhancing their satisfaction, engagement, and overall experience with the system.

2. “the users are involved in the design and development phases” (Tosi, 2020, p. 50)

A key aspect of HCD is the emphasis on empathy with users. Designers are encouraged to deeply understand the emotional connections of the people they are designing for. This often involves engaging directly with users through methods like interviews, observations, and other ethnographic approaches to gather insights about their needs and challenges (Tosi, 2020, p. 69). Tosi (2020) writes how this is a necessity due to how people are individuals, and therefore their needs cannot be generalized.

> "People's needs, therefore, cannot be generalised, except where they pertain to the fundamental needs related to the basic characteristics of human beings [...] The needs analysis is only valid if it is conducted via structured and verifiable investigative methods, which consider the characteristics, attitudes and abilities of a specific group of people under consideration and the activities and context for which the product has been designed or is used." (Tosi, 2020, p. 69)

By building empathy and a deeper understanding of the end-users' needs, designers can ensure that the product addresses real-world issues and aligns with users' expectations (Tosi, 2020, p. 75).

3. "the design is led and refined by human-centred evaluation" (Tosi, 2020, p. 50)

This suggests that the design process should be guided and improved through evaluations that focus on the users and their experiences. This principle suggests that throughout the design and development phases, continuous assessment of how the design meets user needs and expectations is crucial. *Human-centred evaluation* involves gathering feedback from users regarding their interactions with the product or system, identifying usability issues, and understanding their emotional and cognitive responses.

4. "the process is iterative" (Tosi, 2020, p. 50)

The design process itself is iterative, meaning that designers continuously test and refine their ideas based on user feedback. As demonstrated in Sebastian Büttner and Carsten Röcker's (2016) field report, the cycle of prototyping, testing, and redesigning ensures that the product evolves in response to user needs and remains user-friendly and effective (p. 4). Büttner and Röcker were able to apply HCD methodology on a field of work that focuses on functional requirements while neglecting the usability and user experience of products. They found that the iterative process allowed by involving users at every stage of the design, designers are able to make informed decisions that lead to more successful and impactful outcomes (Büttner & Röcker, 2016, p. 1).

5. "the design addresses the entire user experience" (Tosi, 2020, p. 50)

HCD also takes a holistic, systematic approach to design, recognizing that products exist within a broader context. This context includes social, technological, and environmental factors that influence how users interact with products:

> "the needs of each individual also vary based on the reason for which the product is used, contributory factors (haste, fatigue, distraction etc.) and, finally, the expectations the user has for the product, which may relate to its functionality, price, appeal, aesthetics, simplicity and ease of use, etc." (Tosi, 2020, p. 69)

By addressing the entire user experience, designers aim to create products that are not only functional and easy to use but also enjoyable and satisfying. This involves considering how users feel before, during, and after their interaction with the design.

6. "the design team includes multi-disciplinary skills and perspectives" (Tosi, 2020, p. 50)

The final principle emphasizes the importance of having a diverse team with various expertise and viewpoints in the HCD process. This diversity is important as it allows for a further comprehensive understanding of user needs and the complexities of the design challenge itself. By incorporating different disciplines, the team can address various aspects of the user experience during the differing iterative steps of the process.

By considering these six values, HCD helps designers create solutions that are relevant and meaningful to the user (Tosi, 2020, p. 69). Overall, HCD is an inherently empathetic approach to problem-solving and innovation as stated by Andrew Schall (2024) in chapter one of his book *The Persona and Journey Map Playbook*. Here, he outlines ten key principles, of which number one is empathy (p. 2). With this in mind, HCD helps gives us the necessary tools and perspectives when taking a user-centric approach to a process. Its iterative and inclusive nature offers strategies for exploring the users themselves, which aims to conclude in better designs.

4. The Definition of a Game

What are video games? A seemingly simple question that undeniably needs to be asked in order to understand why the media is relevant to what this paper aims to explore. Similar to sports, tabletop games, or gambling, video games fall under the broader category of simply *games*. As such, it is essential to define what elements create a game in its most fundamental sense. This paper uses the definition made by Jesper Juul (2003) in his article *The Game, the Player, the World: Looking for a Heart of Gameness.* Here, he attempts to find a general description and definition of a game by analyzing how other writers write about games. He proposes six features, which he refers to as the *classical game model* (Juul, 2003), leading to the following definition:

> "A game is a rule-based formal system with a variable and quantifiable outcome, where different outcomes are assigned different values, the player exerts effort in order to influence the outcome, the player feels attached to the outcome, and the consequences of the activity are optional and negotiable." (Juul, 2003, p. 35)

The classical game model defines games as a rule-based formal systems with a variable and quantifiable outcome, where the results depend on the player's actions.

Jesper Juul is a Danish game designer, educator, and researcher, renowned for his work in video game studies. He holds a PhD in video game theory from the IT University of Copenhagen and is an associate professor at the Danish Design School. Juul has contributed significantly to the academic field of game design, where his research focuses on the intersection of game rules, narrative, and player experience. Juul is considered a Ludologist; someone who studies games and gaming, especially video games. What makes the books and articles he writes worth focusing on, compared to those of more traditional game designers, is the humanitarian perspective that ludology provides. Ludologists argue that games reveal a great deal about individuals, not just in how they play, but in how they interpret and engage with the experiences games offer. This perspective is particularly relevant when considering video games, which are primarily created for consumers. As commercial products, they must provide value to those who choose to purchase and play them. Consequently, the quality of a game is often viewed as subjective, shaped by each player's unique experiences and interactions with it. This incentivizes developers to create games that appeal to the needs and wants of the consumer, meaning that games are modelled to be something the consumer will enjoy; video games are made to appeal to specific demographics for this reason. From the view of a ludologist, games speak to the undertones of the human psyche, meaning that if we understand why players enjoy games the way they do, we can link emotional and psychological response to the design choices made. This perspective is the reason why this paper uses his writing as a baseline for understanding video games and video game design.

4.1. Understanding the Classical Game Model

To get a better overview of the classical game model, we can condense it into six identifiable aspects based on Juul's (2003) definition, as outlined by Dormans (2012):

1. Games are *rule based,*
2. and have *variable, quantifiable outcomes,*
3. which are affected by the *effort* of the player,
4. and which are assigned different *values*,
5. and to which the player is *emotionally attached*,
6. and which *consequences are negotiable.* (Dormans, 2012, p. 4)

Games operate within a defined set of rules that dictate the permissible actions and structure of play, which is also what enables the creation of game mechanics. These rules ensure that players know the boundaries within which they can act, creating an environment where the outcomes can vary with each play (Juul, 2003, pp. 36-37). As Juul notes, the game's outcome must be variable to be considered a game: "If the outcome of a game is the same every time, it does not qualify as a game" (Juul, 2005, p. 31). The outcome of a game is also quantifiable, meaning it can be measured objectively, often through scores, rankings, or achievements (Juul, 2003, p. 37).

Furthermore, different outcomes are assigned different values. For example, the value of winning may be positive, while losing might be negative (Juul, 2003, p. 37). These outcomes contribute to the game's progression and player engagement, with players striving to achieve higher rankings or specific milestones. The player's involvement is further emphasized by the requirement to exert effort. Effort might involve strategic decision-making, skillful execution, or overcoming challenges within the game, making the game an active experience (Juul, 2003, p. 38). Through this effort, players develop an emotional attachment to the outcomes. Winning often brings feelings of happiness, while losing leads to disappointment. This attachment is a psychological feature of games, rooted in the game contract, where players agree to care about the results, whether the game is a game of skill, chance, or a combination of both (Juul, 2003, p. 38).

Finally, the consequences of a game are negotiable. Players have the option to decide how seriously to take the game, and they can negotiate the significance of the outcome beforehand. For instance, in a casual setting, a game might simply be a form of entertainment, while in a competitive context, real stakes such as bets or consequences may be added (Juul, 2003, pp. 38-39). This flexibility is one of the defining characteristics of games, where consequences are not always fixed and can vary based on the context and the players involved.

The classical game model acts as a foundation, that demonstrates how games, through their structured yet flexible rule sets, create conditions for players to cultivate strategies within constraints. This classical model underpins how fixed rules enable but also limit player agency and unforeseen outcomes. This model and definition is used as a baseline for understanding how video games are constructed, that lets us further explore design approaches from here.

4.2. Game Mechanics: Video Games in the Context of Interaction Design

Fundamentally, video games are a series of interactions. Going down another layer; interaction can be understood as dialogue; the dynamic exchange between a user and a system through a designed interface, with the goal of achieving an outcome. (Sharp et al., 2019). This means interactions are not just about usability but also encompass how users engage with digital and physical interfaces to achieve goals, navigate systems, and shape experiences. Donald Norman (2013) in *The Design of Everyday Things* highlights that good interactions reduce cognitive load, making technology more intuitive, engaging, and accessible. To improve an interaction, a design process employing interaction design can be applied, where the end goal is to create what Jonas Löwgren and Erik Stolterman (2007) refer to as a *digital artifact*; something digital made by humans. Due to the context of this paper, the artifact in question will mostly be considered video games, but using ‘digital artifact’ as a general term allows for a generalization in the knowledge this paper seeks to derive, in order to make it applicable in other contexts.

An interaction in the context of interaction design is not merely an action but a continuous conversation between user and system that shapes user experiences. This continuous conversation can be seen as what Joris Dormans (2012) refers to as a *feedback loop* (p. 19). It is essentially a revolting instance of interactions feeding off other interactions. For example, the player of a video game provides an input via an interactive device, the game system then reacts to these inputs with an output, the player then reacts to the output with another input and so on. This feedback loop then repeats and evolves until either the player decides to quit playing, or the system reaches a state of “game over”, whereafter the player can choose to start the loop anew. Therefore, it can also be seen as the core gameplay loop of a game. Each feedback loop facilitates unique interactions due to the complexity and interplay of the game mechanics, which inherently breeds emergent outcomes that shape future interactions and define the entire experience as a whole.

Video games, as a form of entertainment, fundamentally differ from other interactive systems, such as websites or mobile applications, in that their primary purpose is to provide users with a sense of gratification. This gratification is most commonly derived from the enjoyment of the experience itself, specifically, from the act of having fun during play and not necessarily after the fact. While other types of user interfaces, such as those used in websites or applications, also prioritize user experience, their primary focus is on usability by optimizing ease of use and ensuring user-friendliness. Although these interfaces may provide satisfaction or a general sense of enjoyment, they rarely prioritize "fun" in the same way video games do. The effectiveness of such interfaces largely depends on practical concerns, such as clarity and efficiency, which are essential to achieving a positive interaction. In contrast, the user experience of a video game may be less reliant on these practical considerations, as long as the core gameplay loop remains engaging and fun. In fact, a video game's success can often be attributed more to the enjoyment of the gameplay itself than to the seamlessness of its interface design.

This phenomenon makes interaction design seem less important, however, using interaction design in the context of games can prioritize other aspects than the user interface. Instead, it provides a way to improve the core of what makes games engaging through its game mechanics. As described by Dormans (2012), a *game mechanic* refers to the specific systems derived from the rules of the game, that govern the behavior of elements within a game. These mechanics dictate how players interact with the game world and each other, influencing the overall gameplay experience. Game mechanics can encompass a wide range of functions, including movement, resource management, combat systems, and scoring methods, but together they make up the core identity of a game's gameplay (Dormans, 2012). Game mechanics are often distinguished from the broader concept of game rules. While rules provide the overarching framework for how a game operates, mechanics are the detailed implementations that translate these rules into actionable gameplay (Dormans, 2012). For instance, a rule might state that a player can move a character, while the mechanics specify how far the character can move, the conditions under which movement is allowed, and the effects of that movement on the game state. In the original Super Mario Bros. game, the rules of interacting with an enemy say that touching the enemy means the player character dies and must start over. However, it also states that hitting the enemy on the top as the player is descending will eliminate the enemy and provide points. These rules combine to create the game mechanic of jumping on top of enemies to succeed. The importance of game mechanics lies in their ability to create engaging and dynamic interactions within the game. They are essential for shaping player experiences through the core gameplay loop, and are the core element that makes a game fun, thereby defining the quality of a game.

Overall, interaction design provides a lens through which to understand video games, given that video games are inherently interactive, having players engage directly with systems and interfaces, interaction design allows us to explore how these interactions influence the user's interactive experience. At the core of video games, lies a dynamic relationship between the game and the player. Interaction design focuses on how these systems are structured, how players navigate them, and how the design choices impact engagement, and usability. Video games, as interactive systems, benefit from interaction design by ensuring that players can easily understand the rules of the game, the mechanics at play, and the outputs of their actions.

4.3. Progression and Emergence

Juul (2002) argues in his article *The Open and the Closed: Games of Emergence and Games of Progression*, that most modern video games are comprised of two different ways to present the player with a challenge. One being through *progression*: a more controlled and linear structure, where players encounter predefined challenges in a specific sequence. This is a format often used in story-driven and adventure games, as it allows for a more rigid and controlled gameplay that keeps the narrative flowing as intended by the developers. The other way is *emergence*: a dynamic game structure that arises from a small set of simple rules, creating vast possibilities for gameplay variations. This aligns with Morowitz's (2002) general description of emergence, where, in the context of video games, emergent behavior arises from players interacting with simple rules that allow the player to explore various outcomes rather than being confined to a predetermined one. Generally, emergent games are characterized by high replayability and the development of player-driven strategies, that while unintended by the developers, are often encouraged. Put simply, progression brings predesigned challenges with clear-cut ways for players to progress, whereas emergence brings the possibility of a variation of outcomes based on the rules instated. 'Possibility,' being the key word here, as a game having emergent elements does not mean that there is not an intended way to play. This ultimately means that games are not either fully progressive or emergent, but a mix between the two. An emergent system creates a fluid balance where designers can influence gameplay without dictating every event. This balance, according to Juul (2002), is essential for creating engaging and meaningful game experiences, hence why this balance is important to consider. Observing game history can showcase how emergence is fluid in this way, as games and their development seem to evolve over time. As exemplified by the aforementioned Street Fighter II example; it is not just the players and their strategies that adapt to a game's foundation; the games themselves, and the ways developers approach them, must also shift in response to players to remain relevant.

This is especially prominent in what author Roger Caillois (1972) would call games of Agon, more commonly referred to as competitive games or Player-Vs-Player (PvP) games, where the *meta*, or what the players consider the most optimal strategy, facilitates a frequent need for the game to change. Together with *Agon,* Caillois (1972) also describes his four fundamental categories of play: *Agon, Alea, Mimicry,* and *Ilinx* (p. 18). These provide a systematic classification of games based on the dominant characteristics that define them. Most relevant is Agon and Alea. Agon focuses on competition and rivalry, where players confront each other under equal conditions to demonstrate superiority in a specific skill or attribute (Caillois, 1972, p. 19). This category aligns with what is considered the Player-vs-Player (PvP) genre, which is particularly evident in the modern E-sports scene that has flourished over the past decade. Games like *Counter-Strike* and *League of Legends* embody this category, where competitive multiplayer is the core focus. It encapsules the type of game that is built upon strict rules; games where the designers had clear intentions and made specific rules that would cover most situations. The outcome is expected: there will be a winner and a loser based on whoever is able to best master the rules applied. Alea*,* on the other hand, is about chance and luck, where the outcome is determined by fate rather than player skill, emphasizing a passive attitude of awaiting results (Caillois, 1972, p. 21). While games based purely on Alea, such as dice rolls or roulette, are games of pure gambling, digital games rarely rely solely on chance. Instead, Alea can be tied together with emergence as it can serve as a tool for introducing randomness and unpredictability. It can be used for different intentions like making playable characters randomly trip in a competitive, Agon game like Super Smash Bros. Brawl in order to even the skill-level between players, or as a way of adding excitement by randomly generating playable areas for players to explore in games like Minecraft. As players continuously push the systems of Agon games, new more potent strategies will emerge. Some strategies may be too strong for the current balance of the game; something essential to the level of competitiveness of a competitive game, yet, they may still be fun to perform at a mechanical level. The designer’s job here, is to consider and analyze these strategies, then make changes accordingly, that do not disturb what makes these strategies fun to perform in the first place. This becomes a matter of balancing progression and emergence. A personal favorite example of this phenomenon, once again comes from the Street Fighter franchise. This time, in Street Fighter 5, a character by the name of Dan was added. Dan is typically known for being a joke character, in the sense that he is designed to be weaker, but personality wise, he believes he is the strongest around. Therefore, he makes use of a lot of taunts in his move-set. However, shortly after he was added to the game, players figured out a way to perform an infinite combo by using his taunts to cancel the ending of other moves, the affectionately named “Dan-finite”. This meant, that if a player managed to hit the opponent with the right opener, as long as the player could perform the correct series of button inputs, they could loop this attack and taunt rotation and would win guaranteed. This infinite combo was not easy to perform, so players enjoyed the challenge of learning how to do it, so instead of simply adjusting the moves to make the infinite no longer possible, the developers decided to add an enhanced version of the move used in the combo, that would randomly come out when performing it. This effectively made Dan a stronger character, but the catch is, that the new random version of the move would knock the opponent to the ground, meaning the combo would end. By doing this, the developers managed to maintain an emergent strategy that players enjoyed performing, while reducing the frustration of being on the receiving end. Phenomena like these showcase the importance of understanding and working with emergence while collaborating with the end-users during a design process. Depending on the desired outcome, it is up to the designer to shape the basis of the user's journey by finding a balance between progression and emergence. Using this for the purpose of game design, it can be said that elements of Alea are necessitated as a reaction to the rules of Agon. Translated, this relates to how emergent behavior is born from players overcoming the rules of a game. This idea lends itself well to an iterative design process, as it provides a perspective from which to peak into how future iterations of a design might look. Allowing and embracing emergent behavior in these scenarios, gives opportunities to analyze both how and, more importantly, why players perform the interactions they do, in order to ensure a design iterates in the right direction.

4.4. Player Agency – Consequences and Emotional Investment

When a player decides they want to play a game, they enter into an agreement with the developers, that they will continue playing the game as long as they feel sufficiently engaged. If they do not feel that the experience is worth their time, a player can simply choose to stop playing. Unlike other digital artifacts that aim to provide a service, video games are negotiable, as stated by Juul's (2003) classic game model. As previously stated, video games need to be fun. Therefore, in order to successfully utilize video games as a means of analysis, we need to understand what engages players to keep playing. One approach is to focus on the level of *agency* (Wardrip-Fruin et al., 2009). Wardrip-Fruin, Mateas, Dow, and Sali (2009) define agency as a phenomenon that arises from the interaction between the player and the game, occurring when the actions a player wants to take are both possible within the game and supported by its underlying computational model. Dormans (2012) highlights this by linking the rules of a game directly to agency, stating that "the rules form an interface with the fictional world, and it is through rules that players can affect that world; game rules create agency" (Dormans, 2012, p. 27).

The significance of agency within video games extends beyond merely allowing the player to make choices; it is also central to the way players form emotional connections with the game world. This relationship between agency and emotional engagement is directly tied to the consequences of a player's actions. As discussed earlier, the rules of a game form a stable framework within which the fiction operates, ensuring that players can meaningfully engage with the world despite its fictional nature. However, if the rules fail to provide meaningful consequences for player actions, the player's sense of agency, and therefore their emotional investment, diminishes.

This ties into Juul's (2005) concept of "Half-Real," which describes how games simultaneously operate as both real and fictional entities. While the rules provide a structured and predictable foundation for interaction, the fictional world introduces narrative elements that engage players emotionally. This duality is essential for understanding player engagement: if a game's mechanics do not support its fiction, or if the fiction does not effectively leverage the mechanics, the experience may feel disjointed (Juul, 2005). In this way, agency is not just about the ability to make choices but also about how those choices are reinforced within the game's structure.

Dormans (2012) further elaborates on this by exploring the role of emergence in fostering player engagement. In *Engineering Emergence: Applied Theory for Game Design*, Dormans (2012) examines how emergent gameplay arises from the interplay of rules, player agency, and system dynamics. He highlights that emotional attachment to game outcomes is crucial for sustaining player engagement, emphasizing that players must perceive a real possibility of success or failure for emotional investment to take root (Dormans, 2012, pp. 4-5). This is where emergence becomes particularly relevant: without variable (unpredictable) outcomes, the player's attachment to the game diminishes, as there is no sense of unpredictability that can lead to meaningful consequence.

When provoking emergent behavior, a designer can look at how players respond when backed into a corner; when they are on the brink of failure. Juul (2005) notes that "While a clear valorization (goal) and emotional attachment to the outcome afford the player an opportunity to succeed, they also mean that the player can fail miserably" (p. 199). This emotional investment in avoiding failure is a critical driver of emergent behavior, pushing players to experiment with strategies beyond what the designer may have explicitly intended. Juul further explains that “when there is a clear goal, players are more or less forced to focus on optimizing their strategies, but this may run counter to what players want to do” (p. 199). The fear of failure, combined with the presence of a clear goal, encourages players to exploit the game’s rules in creative ways, leading to emergent strategies that can reshape the gameplay experience.

This is particularly relevant when considering the contrast between progression-based and emergent gameplay. In progression-based games, where outcomes are largely predetermined by the developer's narrative structure, the player's sense of agency is often more limited. While they may be able to make choices, those choices usually function within a predefined set of possibilities that lead to specific, scripted outcomes. As a result, player actions often feel justified by the overarching narrative rather than by personal decision-making. A player may feel compelled to carry out a morally questionable act because it is required for progression, but they may not experience the same level of emotional weight that would accompany a similar action in an emergent game.

Emergent gameplay, by contrast, places greater responsibility on the player’s decisions, allowing them to actively shape the world and narrative through their choices. This increased agency fosters a stronger connection between the player and the game world because the consequences of their actions feel more personal and impactful. Unlike in progression-based games, where the fiction can often provide justifications for player actions, emergent games force players to confront the weight of their choices directly. This often leads to greater hesitation, reflection, and ultimately a deeper emotional engagement with the game world.

For example, in a progression-based game like The Last of Us, players must engage in violence to advance the story, and while these actions carry emotional weight within the narrative, they remain fundamentally dictated by the game’s structure. Players may empathize with the characters, but their actions are largely preordained. Conversely, in an emergent game like

Baldur's Gate 3, the decision to steal from a friendly character or attack an NPC is not mandated by the game's progression but is instead entirely dependent on player choice. This distinction makes the consequences feel more tangible because they are not imposed by an external narrative but instead arise organically from player agency. As a result, the emotional attachment to those decisions is heightened, reinforcing the player's sense of responsibility for their actions.

Dormans (2012) highlights that the ability of a game to evoke strong emotional responses depends on how well it integrates consequences into its mechanics. If a game's rules do not effectively communicate the weight of player actions, then the emotional engagement will be diminished. This is why games that emphasize emergent gameplay often feel more immersive; players perceive their choices as meaningful, leading to a heightened sense of responsibility and investment in the game world. Juul (2005) further supports this by emphasizing that consequences drive player engagement, as they create a feedback loop between input and output, making players feel that their actions truly matter within the game.

Ultimately, the strength of a player's emotional attachment to a game is determined by how well the game balances agency and consequence. If a game provides significant agency but fails to reinforce it with meaningful outcomes, the player may feel detached or indifferent. Conversely, if the game effectively integrates consequences into its mechanics, players will develop a stronger connection to the game world, increasing both their engagement and emotional investment. By understanding this dynamic, game designers can craft experiences that not only entertain but also resonate deeply with players, making their choices feel both significant and rewarding.

4.5. Expected Progression and Emergence

The amount of freedom given to users by the relative designers plays a crucial role. This factor influences the balance between progressiveness and emergence; the level of emergence. To frame this idea more clearly, this paper creates its own terms "expected progression" and "expected emergence" based on Juul's (2002) ideas of progression versus emergence in the context of meta-design as discussed by Fischer and Giaccardi (2006). These new terms are used to frame the level of unpredictability the designer or researcher anticipates for in advance. As Gaver et al. (2022) write: “Whether practice-based design research is conducted in an intention-bound or emergent-friendly fashion depends on researchers’ openness to surprise and change, and this may vary in the course of a project” (Gaver et al., 2022, p. 8). It is in the hands of the designer or researcher to facilitate a design environment in which the preferred level of expected emergence and progression can occur. In that sense, the expected emergence and progression can be viewed as two sides of a dynamic scale, where the designer creates interactions that will be placed in either bowl, ultimately shifting the scale in either direction depending on the nature of said interactions. The higher the expected emergence, the more emergent behaviors can be expected from the user and system.

The reason for including this, lies in creating awareness on the side of the designer. Like with the Dan-finite phenomenon presented earlier, arguing that elements of Alea are necessitated as a reaction to the rules of Agon, expected progression and emergence should be balanced in accordance to each other. Consider the example of a standard web browser’s bookmark menu. The default user journey is straightforward: users can add a webpage as an icon on their toolbar, and by clicking on this icon, they are directed to the corresponding webpage for easy access. This represents the expected progression. However, the potential for emergence arises when users begin to explore the hidden affordances within the available interactions. In this case, users are free to customize icons and text. Initially, bookmarks are given standard names and icons. Over time, a user may realize that the text next to the icon takes up unnecessary space and that the icon alone suffices for distinguishing between bookmarks. As a result, the user might decide to remove the text to fit their need. However, when attempting to erase the name entirely, the browser returns an error message, stating that the name is invalid. To work around this, the user opts to rename all the bookmarks to a ‘space’ character, effectively creating the illusion of an empty label. As the user continues to refine their setup, they may wish to categorize their bookmarks. Faced with a lack of obvious features to separate them, the user resorts to creating a few random bookmarks and applies the same technique: removing the text and replacing it with a custom icon that includes a vertical line (see figure 3). These custom icons are then used as separators, successfully organizing the bookmarks into categories. This example demonstrates how the simple rules of the design that shape the intended progression can lead to emergent behaviors as users navigate the available features.

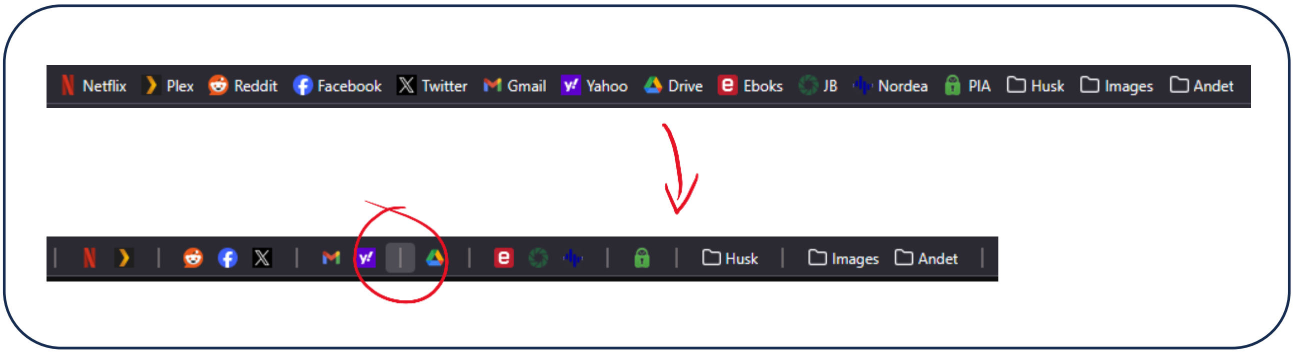

Figure 3 - Firefox Browser Bookmark Example

The potential in unexpected affordances inherent in a design shapes the potential for user-driven innovation and satisfaction. These phenomena are emergent outcomes. Although these customizations are not necessary for the user to complete the intended progressive user journey, they provide flexibility and enhance the user experience. Interestingly, this workaround is not necessary anymore as today, these features, such as text removal and built-in separators are standard in the Firefox web-browser, reflecting how attention to emergent user behavior can drive improvements in design. This once again highlights the importance of paying attention to how emergent user behavior can improve a design, making the relevance for a design approach that tries to capture qualities of emergent outputs more appealing.

4.6. Why Use Video Games for Exploring Emergence?

To summarize, we can ask ourselves "why use video games for exploring emergence?" The answer lies in their capacity to combine rule-based systems, player agency, and dynamic feedback loops in ways that promote emergent behavior.

As discussed, emergent outcomes frequently arise from individual interactions within a larger system; behaviors that were never explicitly planned, yet they are made possible by the underlying rules and structures, as exemplified by interactivity in video games (Juul, 2005). Because video games are so interactive and encourage players to experiment, they are perfect for appropriation. Instead of imposing an established path, designers can help this process along by using meta-design to structure systems that change in response to user input (Fischer & Giaccardi, 2006). Based on previous work by Juul (2003) and Dormans (2012), this paper continued by defining video games more formally as rule-based, outcome-driven systems that emphasize player action and consequence. It became very evident how game mechanics interact with emergent behavior: games facilitate the negotiation of consequences, the generation of measurable outcomes, and the reward of effort. New experiences evolve in games, whether in storylines driven by progression or in systems that allow for open-ended interactions. This shows how games are like living systems, influenced by both design and, more importantly, use.

Ultimately, video games are not just interactive systems; they are emergent systems. They actively provide foundation for interactions without prescribing it. As such, they serve as a practical, observable, and manipulable context for studying emergence in design. The player's journey through a game mirrors the kind of adaptive processes that emergent design seeks to support. This makes video games suitable for understanding and testing emergent methodologies in interaction design and why they were used in this paper.

5. Defining Emergent Design

For the purposes of this paper, we have discussed the idea of 'emergence', but emergent design as a standalone term also needs to be defined in order to ensure a common ground from which to build the design framework this paper attempts to define. Therefore, we can summarize the literary exploration so far to act as a foundation from which this definition can emerge.

So far, we have introduced the concept of emergence and shown how it is fundamental to the adaptive and evolving nature of digital artifacts, as well as how basic rules and interactions can lead to complex outcomes. Moreover, we examined existing methods in interaction design that make use of it. Emergent interfaces (Murray-Browne & Tigas, 2021) and practice-based design research (Gaver et al., 2022) are two examples of innovative approaches to human-computer interaction that challenge the status quo of traditional HCI. These methods propose that users can benefit from more meaningful experiences when designers embrace uncertainty. If these theoretical stances are to be believed, then emergent design has the potential to be more than just a result of iteration; it could be a guiding principle all by itself.

With that, the concept of emergent design fundamentally differs from other traditional design approaches. Traditional designs typically aim to predict and control user interactions, constraining possible outcomes to avoid uncertainty or deviation from predefined paths (Juul, 2005). While progression-based systems deliver a controlled experience crafted by the designer, emergent design deliberately constructs flexible systems that foster unpredictable environments where user interactions determine the outcomes (Juul, 2002). In essence, traditional designs employ a top-down method, where user experience is explicitly dictated by designers, whereas emergent design promotes a bottom-up approach that lets user interactions organically shape the experience.

However, emergent design does not suggest the absence of design structure or intent. Rather, it redefines the designer's role from a strict creator to a facilitator or curator of environments conducive to creative user exploration. This distinction can be seen as the difference between desirable emergent behaviors, such as innovative player strategies, and undesirable emergence, like unintended exploits or broken mechanics that actively halters the user journey.

Furthermore, Dix (2007) argues that systems designed with appropriation in mind can more effectively accommodate diverse user needs and encourage creative problem-solving. Similarly, Fischer and Giaccardi (2006) discuss the concept of meta-design, describing systems intentionally structured to support ongoing user adaptation and innovation. Meta-design advocates creating open-ended environments that empowers users to actively participate in shaping their experiences, thus promoting emergence as a collaborative process between designers and users (Fischer & Giaccardi, 2006). Both appropriation and meta-design reinforce the validity and theoretical grounding of emergent design through human-centered design values, highlighting that fostering user-driven creativity through structured openness is beneficial and practical.

Therefore, in this paper, emergent design is defined as a deliberate design methodology that intentionally encourages unforeseen user behaviors as emergent outcomes, treating unpredicted appropriations as beneficial rather than problematic. Emergent design is therefore about creating interactive systems through co-design that facilitate and embrace unexpected affordances in the form of appropriations, transforming spontaneous user interactions into valuable iterative features (emergent outcomes) of the user experience, to further develop the current system.

6. First Iteration Framework

Having defined emergent design and established its theoretical foundations, the following section introduces the first iteration of a structured framework.

This framework is designed with the intent to support the creation of digital artifacts and systems using emergent design as a main driver for innovation within a human-centered co-design process. While originally developed with video games in mind, the intent is for its principles to extend beyond gaming to other interactive systems, meaning that the aim is to make it applicable to more digital artifacts than just video games.

Furthermore, it is also designed with the purpose of being reiterated upon later in this paper, as currently it will be purely based on theoretical hypothesis based on current literature and concepts showcased so far. Therefore, this initial framework should not be viewed as guideline that is ready to be used, but rather as a method for processing the theory so far discussed as a stepping stone for what is to come. As it stands, four main design dimensions have been found and considered to be the foundation of what could be a framework for working with emergent design: the environmental context, the level of expected emergence, user agency and consequences and designing for underdesign, which will be further explained in this section.

6.1. Initial Framework

<u>The environmental context.</u>

This first guideline is based on Salovaara's (2008) idea that appropriations are situated in each context, and therefore, emergence is also situated in the same manner, as they argue that behavior unfolds in response to local contexts, not just designer intent. Emergence happens through interaction within a social space, whether through in person, real time sharing or asynchronously through means like sharing the effects of your interactions. You cannot design emergence into a system without designing for the ecosystem where emergence becomes visible for others to experience to some degree as well as include affordances for social visibility, feedback, and iteration.

Therefore, the designer must recognize environmental affordances as part of the design system by consider where, when and with whom the system will be applicable.

Strategies:

- Identify where and how the artifact will be used, both socially and materially, by mapping out the environments users will inhabit.
- Recognize the system or artifact as part of a broader ecosystem.
- Design features that allow users to share their interactions so emergent strategies and behavior can spread within the contextual ecosystem.

<u>The level of Expected Emergence</u>

This paper proposes the concepts of Expected Progression and Expected Emergence to describe how much unpredictability a system is designed to accommodate. This is supported by Juul's (2002) concept of progression versus emergence. The idea that no system is purely one or the other justifies treating emergence as a scalable quality and not binary. Due to the nature of emergence, it is hard to control the level of emergence, but that is why it is only the *expected* level of emergence we work with here. It allows designers to manage risk and ambiguity. By calibrating for expected emergence, the emergent outcomes become intentional rather than accidental. Ultimately, the goal is to design a system that allows for user-driven discovery and variation while maintaining enough structure to guide engagement.

Users should have enough control and freedom to influence and create branches withing their experience, leading to diverse outcomes that can shape the direction of the design.

Therefore, the designer should use expected emergence as a dial rather than a binary. It is not only a matter of whether emergence is present or not, it is a matter of how much uncertainty the designer wants to risk for innovation. It can be easy to assume, that the designer knows what is in the user's best interest, but to allow for the iterative process to flourish, it is recommended that the expected emergence is kept as open as possible in the earlier stages to help reveal how it should be adjusted in later iterations. Albeit in design processes with less iterative steps, it becomes harder to keep the expected emergence high as it is riskier when there is less time to adjust for the outcomes later. A designer should keep this in mind when designing for emergence.

Strategies:

- Treat emergence as a spectrum, not a binary.
- Define initial boundaries of user freedom then calibrate by continuously considering the levels of expected progression and emergence. It is recommended to keep the levels of expected emergence broader during earlier iterations then calibrate accordingly.
- Explicitly articulate expected emergent outcomes by trying to predict what might happened if levels of expected emergence are higher or lower, in order to allow for later analysis.

<u>User agency and consequences</u>

Agency is central to emotional investment. Juul (2005) states that outcomes in games matter because they create meaning for players; their actions are seen as intentional and influential. If the system does not acknowledge player decisions, it undermines user agency, which Dormans (2012) links directly to the likelihood of emergent outcomes in games. Emergence often happens when users are pushed to adapt when systems create space for optimization or problem-solving within the rules of the system. By tying meaningful consequences to user actions, a designer can cultivate emotional engagement that increase the chance of users wanting to utilize the level of agency given to them. Agency without consequences lead to meaningless interactions in the same sense that inputs without outputs are useless. Emergence needs this structure that responds to user inputs by creating feedback loops that users care about in order to maintain interest long enough for emergent outcomes to occur.

Therefore, the designer must ensure actions are consequential; feedback must reinforce user agency. This can be done by making sure that there are compelling feedback loops where choices matter, consequences must feel impactful so the player remains engaged through interactions. A designer should also be able to differentiate between desirable and undesirable emergence. By understanding how users interact with and push against system rules, designers can foster emergent behavior that evolves naturally rather than feeling accidental or disruptive; emergent outcomes that increase rather than decrease the user's experience. Strong agency fosters investment by making success, failure, and experimentation personally significant, pushing players to develop strategies beyond what is explicitly designed.

Strategies:

- Ensure users feel responsible for system outcomes by tying outcomes directly to user choices to design feedback loops that reflect and react to user interactions meaningfully.
- Differentiate between desirable and undesirable emergence by keeping track of emotional responses. Do consequences lead to users wanting to overcome the trials they are given, or do they simply lead to uninspiring frustrations?
- Track emergent responses to identify which improve the experience based on emotional investment and how they do so.

Designing for underdesign

Fischer and Giaccardi (2006) describe underdesigning in the context of meta-design as deliberately building systems to be unfinished, allowing users to evolve them in use. Emergent design is impossible in closed systems, so using meta-design to design for appropriation, empowers users to generate new value through emergent outcomes. Emergent systems need structural flexibility, not total openness, and must enable users to appropriate without breaking the system completely.

Therefore, a designer should leave parts of the system open-ended and ambiguous to allow for reconfiguration. They should encourage appropriation by making mechanics that are modular and adjustable; hackable, so to speak. Provide the user with a toolset within the system that is not directly incentivized but instead discoverable, akin to the aforementioned web-browser bookmark example. For example, this could be achieved through modular design or adaptive interfaces that acknowledge and react to user behavior by designing systems that provide contextual feedback and evolve based on user interactions.

Strategies:

- Apply meta-design principles: design systems to be incomplete by design.
- Avoid over-engineering; leave affordances open to interpretation.
- Mark components of the system as flexible or user-editable by prioritizing modularity and adaptability.
- Keep track of specific underdesigned aspects, then study user behavior for signs of creative workarounds during interactions with those aspects.

7. Ethnographic Methodological Approach

The purpose of the following study is to explore the ideas of the first iteration of the proposed emergent design framework through a human-centered, ethnographic field study involving individuals who actively engage with video games. The aim is to rework the current framework by incorporating empirical insights and direct user contact that can guide its future development. The research is situated within the Fighting Game Community Aarhus (FGCA), a local Danish gaming community specializing in the fighting game genre of video games, that meets weekly to play series titles such as *Tekken*, *Street Fighter*, and *Super Smash Bros.*

Following the research of Gaver et al. (1999) on cultural probes in design theory, the following empirical study recognizes the importance of tacit knowledge (Schön, 1992). Gaver et al.'s (1999) goal through their research was to find impressions and personal reflections rather than gather measurable and quantitative data, instead focusing on "inspirational data" (Gaver et al., 1999, p. 25). This was to focus on data that would provide glimpses into participants lives and values that could then inspire speculative design. This perspective was valued during the empirical research, and paves the way for what Schön (1992) refers to as *Tacit Knowledge* (p. 52); the subjective, ambiguous or emotionally charged insights that are difficult to find through structured interviews and similar approaches. It is the knowledge a user has, but cannot explain why they know of it or how to use it. This is what has inspired the chosen methodology used throughout the field study of this paper. The empirical data the research aims to find is inherently tacit, meaning rather than give concrete answers, the data is gathered for later speculation and analysis in order to further explore the possibilities of emergence outcomes. The aim was not to extract explicit user needs, but to become attuned to the situated, affective knowledge embedded in how players interact with games and each other. This is also based on the assumption that emergent behaviors are grounded in the tacit knowledge that causes players to explore and provides motivation.

The next sections present and describe the approach and considerations made in tangent with the ethnographic studies conducted.

7.1. Field Study Preparations – Location, Demographic & Methodology

Overall, the study involved a series of qualitative, semi-structured interviews with small groups of two to three participants, all of whom are active members of FGCA. To support the interviews, a two-month ethnographic study was conducted in which the researcher participated in eight of FGCA's weekly gatherings. Each weekly session lasted approximately five hours. The study is mainly based on ethnographic field methods outlined by Blomberg et al. (2017) to maintain a clear commitment to human-centered research that values real-world interaction as a source of knowledge in both interaction design and game studies.

Before presenting the methods used for the field research and how they were used are discussed; following is an introduction to what Fighting Game Community Aarhus is and what kind of people were present as part of both observations and interviews.

7.1.1. Fighting Game Community Aarhus

Fighting Game Community Aarhus is a local grassroots gaming club centered around fighting games, based in Aarhus, Denmark. In this context, 'grassroots' refers to organizations that are formed and maintained by ordinary members of a community rather than by institutions or corporate entities. A grassroots group like FGCA operates from the bottom up, meaning it is led by participants, for participants, with minimal external oversight or funding. It typically emerges organically around shared interests and relies on volunteer work and collective decision-making to sustain its activities. It serves as both a social and competitive hub for players and those interested in the scene. It is a community driven environment where members regularly gather to play, learn, and discuss fighting games along with other overlapping interests.

FGCA is a rather active community both online and offline. While their main concerns lie with trying to keep the offline, social, and competitive aspirations of 90's arcades alive, they also regularly gather in online meeting rooms to play online.

The main part of what it means to be a member of FGCA is showing up at their weekly gatherings called *locals*, where they meet and play for five hours at a time either to improve or simply to hang out. Other than these weekly locals, they also host monthly tournaments locally referred to as *monthlies*. These are more serious and requires and entrance fee and participation fee if people want to play in game brackets. During monthlies, members of FGCA gather to test their skills as money is now on the line; focusing more on the competitive aspect while still fundamentally being about the community.

FGCA can be seen as a local manifestation of the global Fighting Game Community (FGC). The FGC is not a formalized institution but rather a loose, global collection of communities organized around competitive fighting games. Although FGCA is part of the larger framework that is the FGC, they are still situated as a reflective and generative space, as it inherits overarching, global trends from other corners of the FGC, such as general jargon, strategies, meta-games, interests, etc., while maintaining its own identity through its own members preferences to create a variation of the community.

Figure 4 - Image of FGCA Local

7.1.2. The People

The demographic of the member of FGCA is surprisingly diverse. When looking at the age groups of other competitive gaming environments you would usually encounter a younger demographic. Activision's *Call of Duty League* reports an average competitor age of around 22, while Blizzard's *Overwatch League* averages about 20 years old for its players (Lee, 2022).

However, while spending time at the FGCA locals, it was much more common to encounter players between the ages of 25 and 40. Quite a higher average, although from what the people had to say, they attributed the higher ages to the fighting game legacy being much older than most other competitive scenes. As mentioned, FGCA tries to keep arcade culture alive, and are rather proud of the fighting game community's legacy and what it means to identify as a fighting game player. Generally, even newer player, took pride in being part of the community and not just their skills. Based on observations, FGCA is a community of highly enthusiastic people driven by a love for their community with fighting games as the core.

In regards to diversity other than age, FGCA does represent more than just the average Dane. FGCA is of course located in Aarhus, Denmark; a Nordic country with a majority of Caucasian citizens. Although the observations show many Danish people attending the locals, there are also many players representing other parts of the world. Looking at the FGCA monthly tournament held in April of 2025, it can be observed that just the players placing within the top eight players of one of the game tournaments, included people representing Togo, the Democratic Republic of Congo, South Africa, Australia and Germany (see figure 5). Other than that, the researcher also encountered people representing Thailand, Korea, Poland, the United States of America, Spain and even more.

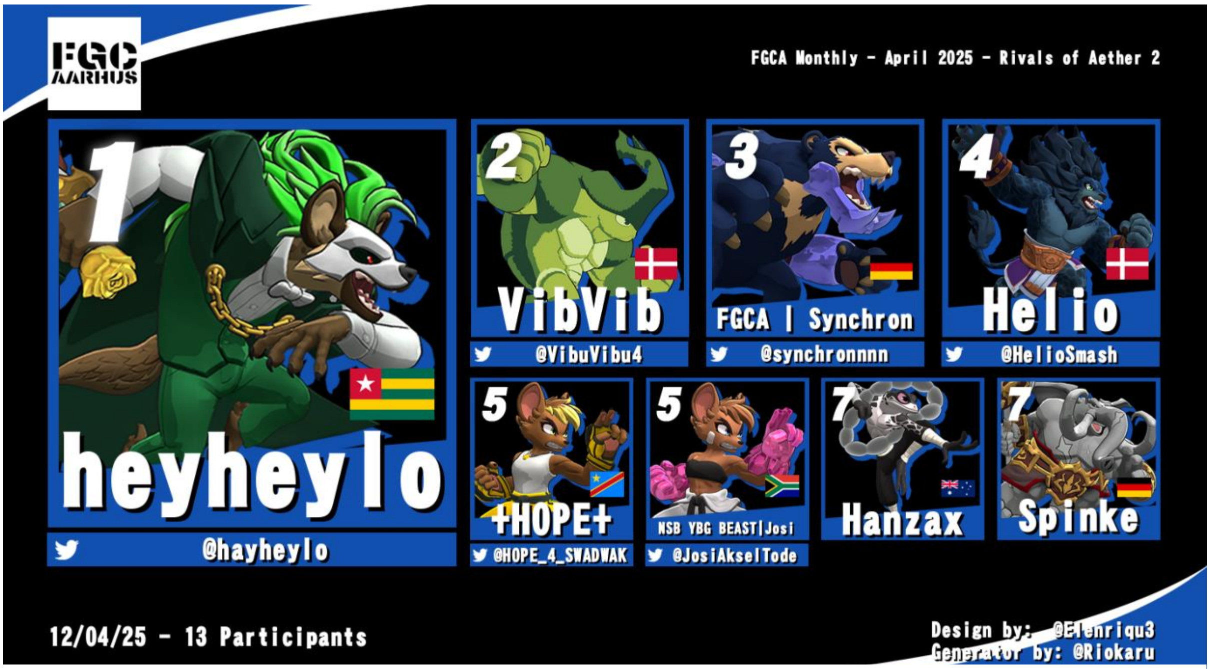

Figure 5 - Rivals of Aether 2 FGCA Monthly Top 8 Graphic (x.com/FGCAarhus)

The generational, cultural and general diversity present, not only enriches the social fabric of FGCA, but also contributes to a larger variety in mindsets and personalities, strategies and approaches to the games and as a result, FGCA is an environment that celebrates diversity and resists homogeneity by prioritizing a shared ethos of love and enthusiasm for community grounded in passion for a common ground.

Figure 6 - Players Playing Super Smash Bros. Ultimate at FGCA

7.2. Ethnographic Observations

Understanding the context of what FGCA is and who they are, we can begin discussing the methodological approach and considerations taken in regard to the ethnographic study of this paper. Starting with the approach to conducting on-site observations.

As it stands, FGCA serves as a social space where knowledge is shared and the community evolves. This makes it a valuable setting for observing emergent behavior in interactive systems. Fighting games are particularly relevant due to their systemic complexity and history of player-driven discovery. As previously discussed, fighting games are inherently deterministic in its game mechanics, which means that players are forced to circumvent the strict mechanics using emergent strategies to keep the competitive interest in the games alive. Considering that there exists a multitude of different fighting game communities all over the world, this goes to show that emergence can still be found even in strict systems. It goes to show, how interactive systems are never 'solved,' if users are sufficiently engaged, as they will then find emergent strategies that further shape the interactions and experiences a system can provide. FGCA provides an accessible and grounded example of this process, offering a space to observe how emergent strategies develops and is sustained within a community that deals with interactive digital artifacts. The community's informal structure, evolving practices, and shared traditions may provide further insight into how players appropriate and reinterpret game systems.

The researcher took on the role of *participant observer* (Blomberg et al., 2017, p. 131), meaning their intention was to immerse themselves within the community by engaging in casual play, informal discussions, and observation of the community's social and gameplay dynamics. This immersive method was chosen to capture behavior and interactions in their natural context, in line with the principles of human-centered design. As noted by Blomberg et al. (2017), removing behaviors from their social context risks altering them in ways that may obscure their meaning. Ethnographic observations address this by embedding the researcher within the environment being studied, allowing for a richer and more accurate understanding of participant practices. FGCA members' perspectives and actions offer a form of participatory feedback that informs the framework's development. Their experiences help challenge and refine the assumptions underlying emergent design by showing how it plays out in actual use. In order to maintain a believable role as a participant observer, the researcher would use their connections to the organizers to send out a message to all participating member that would explain the situation and how the researcher would take on the role of an outsider. In essence, they were briefed and asked to assume the researcher was a new member on a trial, who were considering joining as they were curious about the games played. It stated that the researcher would be walking around with a pen and paper and other remedies, which would indicate that they were 'in character', so to speak. Using simple props to indicate the current role of the researcher helped smooth over awkward greetings and explanations by letting those the researcher interacted with during participating observations enter into a non-verbal agreement, that they would act like the researcher was a curious and fresh member being inquisitive. It established a simple yet effective roleplay, that could be toggled on and off if needed.

A caveat to this approach, is that many of the observations made could not be directly recorded, as many were either experienced with, or, came through casual conversations with a variety of people attending the weekly gatherings. Therefore, unless otherwise specified, it should be assumed that presented data from specifically the field observations were noted by the researcher and then later processed as they are presented in the later analysis chapter.

Furthermore, as noted, gatherings at FGCA bring with it a wide variety of different people, and as only one researcher worked on this, it means that they could only focus on one thing at a time. It was undesired to spread the researcher too thin, as the intend was to immerse them into the context as much as possible. An attempt to remedy this, comes from the number of times the gatherings were attended, making up for potentially missed moments by allowing for more moments to occur over a longer span of time and more visits.

7.3. The Interviews

In order to explore deeper, qualitative, semi-structured interviews were conducted. This was to allow for better contextualization of the participant's decisions and commitments. To do this, the participants were taken aside to perform qualitative interview with the researcher within groups of two-three people, based on a semi-structured approach (Tracy, 2020). This method allows the researcher to guide the conversation while maintaining the flexibility to explore new insights as they arise during the discussions (Sharp et al., 2019, pp. 269-270). The semi-structured interview format is especially valuable when the goal is to generate new perspectives or understand poorly defined phenomena, as they are "meant to stimulate discussion rather than dictate it" (Tracy, 2020, p. 157). By encouraging detailed responses and allowing room for the participant to express themselves, this approach uncovers not just what players think, but also how and why they think it, opening up for deeper insights about their subjective experiences (Tracy, 2020).

In the context of HCD, semi-structured interviews emphasize empathy and helps to better understand the interviewee's perspective. By using semi-structured interviews, the researcher can uncover the lived experiences of players, allowing for a better understanding of their motivations and frustrations. Tracy (2020) highlights the importance of semi-structured interviews in obtaining rich data that provides a nuanced understanding of complex topics. In the case of video games, where player behaviors, motivations, and emotional investment are central to the gameplay experience, this method allows for the exploration of how players engage with games but also why they make certain decisions within the game's framework. By probing deeper into players' individual experiences, the researcher can better understand the ways in which emergent gameplay unfolds and how game designs can adapt to foster more engaging, personalized experiences, by coupling the data with the observational data also gathered. Furthermore, the flexibility of semi-structured interviews enables researchers to also gather supplementary structured insights based on predefined questions, like quantitative data on who the participants are, their interests, favorite games, etc. However, as Tracy (2020) points out, it is critical to avoid leading questions, which could skew the data and compromise the integrity of the findings. By employing this qualitative method, the researcher can draw valuable insights from players that are directly applicable to the development of human-centered, user-driven designs.

7.3.1. The Interview Participants

Among the wide variety of people at FGCA, some individuals naturally had to be picked out to conduct the supplementary interviews. A selection process was conducted to ensure, that no one felt pressured into a situation they were not comfortable with. Firstly, an open invitation was sent out on the public Discord server associated with FGCA, that generally explained that the researcher was looking for interview volunteers. Those who showed interest in participating were then given more information about the process. They were told that the interviews would be carried out as conversations among their peers through a group chat that would take around 45 minutes to complete. The interviews would be conducted on location at the FGCA venue within the same timeframe they would otherwise already be attending the casual, weekly gatherings. Therefore, they would not need to make extraordinary travels or manage their time differently, making it easy for them to step aside for the duration of the interview. They were told, that we would be discussing fighting games and games in general in the context of using the data for writing an academic paper.

In the end, five individuals volunteered and were divided randomly into two groups based on their availability. The group interview setting was believed to be beneficial, as it could allow participants to communicate with each other and thereby feel more comfortable as well as afford inspiration for tacit answers that the interviewer would have a harder time approaching. By also structuring the interviews around existing social dynamics and minimizing disruption to their routines, the study tried to ensure both comfort and authenticity in the data collection process.

Group one included:

- Anders: a 20-year-old male student from Skanderborg. A local competitive player whose main game is Street Fighter 6, although also plays other fighting games like *Multiversus, Dragon Ball FighterZ* and *Guilty Gear: Strive*, as well as other video games casually such as *Team Fortress 2* and *Bloons TD 6.*
- Michelle: a 30-year-old female student from Viborg living in Aarhus with roots in Thailand. Better known as *Farcai* in the community, she is part of the board of volunteers who run and manage FGCA and its events as well as being one of the original founders of FGCA. She does not consider herself a heavy competitive player; usually preferring more chill experiences, but still indulges to stay up to date and because she enjoys being a part of what is the core activity of the community.
- Schadrac: a 24-year-old male student from Vejle with roots in Congo, he is a competitive Super Smash Bros. Ultimate player. He is an active tournament goer and actively travels outside Denmark to compete under the gamertag *Shy* or *+HOPE+* interchangeably.

Group two included:

- Daniel: a half Danish/half English 26-year-old male student living in Aarhus with a technical background in programming and engineering. They recently decided to help
- Malte: a 27-year-old from Tønder living in Aarhus who just finished their education as a pedagogue.

The participants were demographically diverse, able to offer diverse perspectives from both casual and competitive perspectives, some having organizational involvement, and varied game preferences. Their combined experiences help provide deeper insights into the observations made by the researcher.

7.4. Thematic Analysis

Following the field study and interviews, a thematic coding and analysis was employed, following the guidelines and principles outlined by Braun and Clarke (2006). Thematic analysis is a flexible yet systematic method for identifying, analyzing, and reporting patterns within qualitative data. It allows for rich, detailed, and complex descriptions of data while accommodating various theoretical frameworks (Braun & Clarke, 2006).

The approach to thematic analysis in this thesis was inductive and data-driven (Braun & Clarke, 2006), meaning themes were closely linked to the data themselves rather than explicitly driven by existing theory. This inductive approach facilitated the exploration of player experiences within the observed community, enabling the researcher to generate themes directly informed by the participants' responses and interactions observed (see appendix, pp. 43-50). That is not to say, that existing theory is not relevant, as it guided the initial context of both interviews and observations, but will also become more relevant later as the results of the coding is further discussed.

The process of thematic analysis undertaken followed the six-phase model described by Braun and Clarke (2006):

1. Familiarization with Data: Interview transcripts and observation notes were thoroughly read to ensure familiarity with the collected material.
2. Generating Initial Codes: Data segments that appeared relevant to the research question were systematically coded.
3. Searching for Themes: Codes were grouped into broader potential themes.
4. Reviewing Themes: Potential themes were reviewed and refined to find coherence within each theme.
5. Defining and Naming Themes: Each theme was further defined, ensuring each theme had a distinct focus and meaningful contribution to the research objectives.
6. Producing the Report: A final analytic writeup was produced, weaving together themes and data extracts into a coherent report.

The results of these step are what will be showcased and discussed in the upcoming chapter on the ethnographic findings, to create a basis for the framework and arguments presented.

7.5. Researcher Bias

Before moving on to findings, a disclaimer should be given, that the researcher does have previous experiences with FGCA as they have been an active member for a while beforehand. Although this adds a level of bias, it is not impossible to work around as long as the researcher is aware of their personal bias. By engaging in reflective practice to withhold personal biases, the researcher can navigate the tension between having insider knowledge and analytical distance, allowing them to draw on the benefits of being accustomed while mitigating risks of overfamiliarity:

Firstly, the insider status is what lead to the immediate benefit of the collaboration being able to happen in the first place. It made it possible to access informal and situated practices that may not have been available to an external researcher, therefore allowing for a richer and more natural contextual understanding, particularly in identifying tacit social interactions (Blomberg et al., 2017). The pre-existing relationships also helps reduce the social gap between researcher and participants, as it encourages more candid responses during conversation or interviews. The present trust, the researcher has developed over time within the community, facilitates more open dialogue in interviews, enabling participants to reflect on their behaviors and motivations with greater nuance as they could feel ensured that the researcher did not have malicious intentions. This will likely improve the depth of the qualitative data the study attempts to gather. The core of a tightly knitted community is likely to be harder to access, so from a HCD perspective, the insider status lets the researcher skip directly to the point where they can co-participate with the users and players directly, rather than having to fill in gaps from a detached standpoint. The researcher was able to analyze the community and systems from within from the start. This also means, that the researcher would be more attuned to subtle and tacit dynamics that would otherwise be brushed aside as inside jokes or community memes that would require too much secondary research to begin to understand. In short; it means that the researcher will be able to catch on to otherwise tacit expressions, behaviors, or abstractions. It gives access to many more aspects an outsider might accidentally overlook or misinterpret.

While the researcher's status as an existing member of FGCA introduces a potential source of bias, it also affords the unique methodological advantages such as those just outlined: greater access to tacit knowledge, increased participant openness, and a more situated understanding of emergent dynamics. These advantages were leveraged intentionally, while attempting to mitigate interpretive bias.

8. Field Research Findings & Empirical Analysis

Having now completed the formal interviews; we can now begin to look at the data provided. Although the interviews were finished, the researcher would continue to officially be present and observe at the weekly gatherings for the remainder of the officially allotted two months' worth of locals.

It should already be revealed now, that thematic coding showed three particularly significant elements that shape how this paper perceives emergent outcomes: *Context, Identity* and *Temporality* (see appendix, p. 43-45). What connects these, is that they seem to structure the tacit patterns of engagement that lead to emergent outcomes. Thereby making them what can be described as carriers of tacit knowledge; conditions that implicitly contain information that is not formally expressed or hard to verbally communicate with those outside a given context.

The following section aims to present these three carriers of tacit knowledge to analyze the findings associated with them from the conducted field study, including both the observational data and the insights gathered from the interviews.

8.1. Contextual Emergence

Context refers to the specific environments in which interactions occur. Emergent outcomes are not simply reducible to system rules or developer intention as one might initially assume. As Lucy Suchman (1987) argues, action is always situated and it unfolds in response to the specific material and social context of the moment. Plans, including design plans, may guide behavior, but they cannot account for how users actually act.

This section argues that contexts enable, constrain, and co-produce situated emergent outcomes and are therefore vital when designing for emergence. Rather than seeing emergence as 'content' that designers place into a system, this paper argues that it is a phenomenon of context; a relationship between system, player, and environment.

8.1.1. The Role of a Community

Spending time with FGCA showed that emergent behaviors are inseparable from their social and spatial contexts. In particular, community plays a central role in how emergent outcomes as strategies are discovered and developed. These practices are not developed in isolation, but through shared observation and social feedback based on the surrounding environment, matching the arguments of Salovaara (2008).

One example of this comes from Schadrac, who spoke of having had developed a new technique in *Super Smash Bros. 4* which he dubbed “Shy Attack Cancelling” (see appendix, p. 8). While according to themselves, the technique itself is competitively unviable, but what makes it matter is how it emerged: through experimentation, demonstration, and recognition from others in the community. As he explained: “jeg kaldte det Shy Attack canceling, for det var sådan, at nu fandt jeg ud af det, så det er min ting, så jeg satte mit navn på det.” (see appendix, p. 8) Here, discovery and identity are fused. Their motivation was not only competitive viability but it was social visibility, pride and ownership amongst their peers. Schadrac was not just performing their technique as a showcase of skill, rather they were authoring it; staking a claim and proudly embedding it into the social context where it can be recognized. This aligns with Salovaara’s (2008) observation that appropriation is often a community-driven process. Novel uses of technology tend to spread when they are shared and adopted by others (Salovaara, 2008). Communities like FGCA act as incubators for these discoveries as they provide a reason for wanting one’s identity recognized. Communities develop cultures of participation through shared vocabularies or jargon that describe new tactics and styles (Salovaara, 2008, p. 223), such as Shadrac’s example and other specific combo names or memes that relate to happenings or people inside given communities. Moreover, visibility within the community inspires further experimentation. When players witness someone like Schadrac introduce a new move and receive recognition for it, it motivates others to explore the system in similar ways. Salovaara (2008) describes this as appropriation by inspiration, a social feedback loop where one player’s discovery prompts others to reconfigure their own practices (p. 223), thereby prompting further innovation that can be used in later iterations of a design.

From a design perspective, this highlights the importance of supporting community-level interaction. Salovaara (2008) suggests that designers can facilitate appropriation by building tools for sharing, visualization, and communication, enabling users to externalize their discoveries and help others replicate or build on them (p. 223). Traditionally this might be done through functionalities that allow for communication between members. For an emergent design methodology, this means that designing for emergence is also designing for social diffusion as the system should not just allow new behaviors to arise, but support their movement through a user community.

In short, communities like FGCA are not passive sites where emergent strategies simply appear, but they are active agents in the production and circulation of emergence. These sorts of contexts can vary in scale, but the high variance in unique identities within the context does increase the chances of emergent mutations. What begins as an individual exploration becomes a shared innovation, shaped by context and collective framing. Emergence, in this sense, is not just unexpected behavior but behavior that becomes meaningful through community response.

Figure 7 - Players Playing Tekken 8 at FGCA

8.1.2. The Effects of Offline Play

One of the first questions various community members were asked during the visits at FGCA was simply ‘why are you here?’ Not to seem snarky, but due to a genuine interest in understand why they were not just playing from home. This comes from the general observations of e-sports and competitive gaming seen nowadays, where most competition is done online to some extent. However, as interviewee Anders puts it: “man kan ikke rigtig have den rigtige vibe, føler jeg, når man spiller med folk online” (see appendix, p. 6). He emphasizes that co-located play first and foremost just more right; it feels better. While online infrastructure has improved significantly over the past decade, the players at FGCA still overwhelmingly choose to meet physically when possible. One community member described it to the researcher as being an attempt at keeping the spirit of game arcades alive. The choice is not simply a technical preference of avoiding dealing with latency or netcode, though these remain factors, but about the situated experience of play. For many, the offline context affords a different mode of engagement, one that includes instant reactions through a social atmosphere, and real-time feedback during gameplay. One thing that was quite apparent was the high spirits during the weekly gatherings; the vibes were high, so to speak. But these ‘vibes’ are more than just a mood, it represents the framing of the context so that it may afford certain kinds of expressions and experimentations fueled by the people, users play with, whom encourage cooperative play. The emphasis on offline play at FGCA shows a deeper relational orientation: players do not just play the game; they play *with* each other, despite playing *against* each other to win. Although the goal of the game is to win, the goal given the context may differ greatly. Like Michelle, who mentioned enjoying playing competitively, but for the sake of community growth rather than to win and be a high-ranking player (see appendix, p. 2). In these offline sessions, shared physical space becomes a co-author of the experience: screen visibility, ambient noise, and social cues all contribute to how gameplay unfolds. In this sense, emergence is about how the shared space of a given context shapes what emergent outcomes are possible.

From a design perspective, this illustrates that the context of play is not incidental but constitutive. Systems designed for emergent interaction must consider not only what is possible in code; what is stated by the rules of the game, but also what is afforded or limited by the spaces and social setups in which play happens. FGCA's preference for offline play exemplifies this: the very conditions under which emergence becomes possible are shaped by where and how players engage with the system.

Figure 8 - Competitive Players Competing in Tekken 8 Tournament

8.2. Identity, Agency & Consequences

Identity refers to how players or users change their own practices and see themselves in relation to the relevant digital artifact and the community surrounding. Rather than treating identity as a fixed trait, the field research showed that it is rather something enacted and negotiated as interactions play out. Through interactions with interactive systems such as video games, players continually form and reshape who they are as players in relation to the game. The unexpected and improvised aspects of interactions often become center for expressing a personal sense of difference or belonging. In this sense, identity is not simply a byproduct of the overall interaction but a condition for it as users bring expectations, preferences, and histories with them. What makes the user unique as an individual informs how they interpret and respond to the challenges they face.

This section argues that identity is a structuring force in emergent play, influencing how players navigate affordances, adopt strategies, and articulate their presence within the system. Personalities become variables that affect the outcome of emergent phenomenon.

8.2.1. Genre Affiliation and User Expression

Player identity within the FGCA seems to go hand in hand with genre affiliation. While the overarching category players associate with is fighting games, the community is internally divided into sub-genres of which they associate themselves with. In the case of FGCA, it is primarily 2D fighters (Street Fighter, Guilty Gear, etc.), 3D fighters (Tekken, Virtua Fighter, etc.), and platform fighters (Smash Bros., Rivals of Aether, etc.) (Core-A Gaming, 2024). These preferences are not purely mechanical, rather they represent how players articulate their identities within the social field of the community.

From both observations and interviews, genre distinctions repeatedly surfaced as meaningful. Everyone considered themselves to have a 'main game,' but made it apparent that they were generally aware of other sub-genres and games as well. For example, Michelle identifies as a Tekken player, but emphasizes that even within competitive games, her approach is defined more by 'hygge' than competition: "Jeg er en hygge spiller [...] jeg spiller mere for fællesskabet, end jeg gør for det competitive" (see appendix, p. 2). This contrasts with Schadrac, whose primary identity revolves around competition and is considered a top player in his game's region: "Jeg spiller for det meste Super Smash Bros. Ultimate [...] så længe jeg kan spille mod nogen andre, så må det være, at jeg læner mig mest mod af" (see appendix, p. 3). These genre preferences correspond not only to gameplay mechanics, but also to different identity performances. For instance, Smash players like Schadrac described the extensive steps needed to create a competitive setup that requires removing stages, disabling items, and generally enforcing community-wide house rules, which contrasts with the streamlined start-up of more modern fighters like Tekken. As he puts it: "Hvis Michelle vil spille Tekken, så går hun bare ind, hun vælger sig en figure, hun vælger en bane, hun trykker start. Jeg skal gå ind, jeg skal fjerne 88 baner" (see appendix, p. 6). This labor of preparation becomes part of the identity performance itself which signals dedication, knowledge, and alignment with a specific subculture.

Notably, genre-based identity does not necessarily result in rigid boundaries. While the physical setup of FGCA gatherings often have people play in groups representing their sub-genre, interactions between subgroups were marked by curiosity and playful comparison. Michelle, despite primarily playing Tekken, references modding in Smash and acknowledges cross-genre learning and exchange, showing awareness of the other sub-genres' norms. Similarly, several players at the venue described using combos or techniques from one game in another, such as Michelle transferring combos from the Persona 4 fighting game into Dragon Ball FighterZ (see appendix, p. 11), or Tekken habits into Street Fighter, for better or for worse, that further complicates the idea of fixed genre allegiance. However, this is from the context of fighting games and may seem too narrow to be useful. Here, the important thing to notice is the allegiance users make towards the systems they engage with. Being a Smash Bros. player who tries to play a more traditional 2D fighter is likely to interpret the system's affordances differently, the same way an Apple Mac user might approach using a Windows machine differently. Or how someone who identifies as their favorite fruit being a banana, might approach eating an apple for the first time differently. They might look for a way to peel the apple instinctively. While peeling an apple because you are used to eating bananas is not disallowed, it is also not encouraged by the 'intent' of the fruit. This is inherently an emergent outcome based on identity formed from previous assosiations. This fruit example might seem excentric, however it helps translate the thought process that has led to thinking why genre affiliation matters when analyzing how users interact with different systems than what they are used to. FGCA specifically, provided a lens through which the researcher could observe how successful cross-pollination between genre affiliations provide users with the ability to approach other systems with toolkits from other systems, that lets them prod and understand system affordances differently, to provoke emergent outcomes as they continue. When players from one affiliation jumped over to try someone else's game, it was apparent that the way they spoke about the interactions and how they instinctively moved their character on screen was different from how the more seasoned player was interacting. Schadrac exemplifies this from the perspective of the highly competitive player he is, using a metaphor: "en golfspiller, der bruger sin putter meget længere væk end normalt. Folk vil sige sådan, at det ikke giver nogen mening, men hvis det er hvad, der virker for ham, så er det sådan, han expresser sig selv, når han spiller golf" (see appendix, p. 16). Through this, he also highlights that he finds personal expression important. Watching others express themselves according to their identity can be inspiring for those around them. How a player expresses themselves is dependent on the personal history they have in relation to what they associate with. The way these observed players expressed themselves through their interactions lead to new conversations between the players; the 2D fighting game player often having to translate their jargon to something the Smash player better associated with. It also meant, that the 2D player had to adapt to how the Smash player played, meaning it led to new emergent strategies, that would otherwise not manifest, if the 2D player's standards had not been challenged by another affiliating player. This all goes to show the value of letting users express themselves however it pleases their identity based on affiliation.

Moreover, FGCA's relatively loose genre boundaries seems unique at least in the Danish fighting game scene, as it contrasts with other communities. A visiting player from the Copenhagen fighting game community commented on how they tended to remain genre-segregated in Copenhagen, while Aarhus fostered more crossover and shared interest, at least according to their personal experiences. This supports the idea that identity affiliation is contextually based within socially constructed contexts, rather than inherent or static.

Genre functions in FGCA are not gameplay categories, but as a social and symbolic scaffolding through which identity is enacted. The observations show that genre preferences inform how players present themselves, relate to others, and navigate the community, making genre affiliation an axis of player identity and how they interact with systems.

8.2.2. Performing Perception

Many of the emergent behaviors observed at FGCA were not developed in isolation, but rather in public, competitive settings. Gameplay sessions unfolded in front of small audiences, whether that meant two or three people casually watching over the shoulder or a more focused crowd during a tournament bracket. These moments revealed how players adapted their playstyles in response to knowing they were being observed by others.

In these contexts, certain behaviors emerged that would otherwise defy what would be considered good competitive logic. Players would occasionally attempt unorthodox combos or risky plays, not because they were optimal, but because they were impressive or funny in the moment. A match can be seen as a platform for personal expression, a way to be seen doing something expressive.

This performative layer of strategy aligns with Dalsgaard and Hansen's (2008) concept of *performing perception*, where users engage with interactive systems both as operators and performers by interfacing not just with the system, but with an audience as well. Strategies change significantly depending on whether the context is casual or competitive and these changes are not purely technical but deeply performative. Considering performing perception, players at FGCA can be understood as simultaneously operators of the game, performers of their own skill and persona, and spectators of both their own and others' actions. This dynamic shifts based on context, particularly in the presence of an audience. This becomes apparent as a form of expressive emergence, in which players engage in strategic actions that reflect their identity and social aspirations. In a private context, players may feel free to explore the system with looser constraints. Emergence here is driven by aspirations of curiosity or novelty. But as observed at FGCA, in public, particularly in competitive matches, players perform within expectations by choosing safer, more explored strategies, aspiring to conform to the meta or demonstrate mastery. As a performer, there now exists a layer of consequences based in social perception other than simply losing the match and being out of the tournament. As a performing competitor, the player has an identity that they wish to maintain or keep building on; a persona with a style in the eyes of the community, of which the spectators in the current context have assumptions and expectations about. This creates a layer of urgency that makes users more likely to choose safer interactions within the system in order to not lose face in the moment while performing. Here we also see that a looser, more casual environment with less urgency might more easily foster emergence in-the-moment, as failure is less of a fear, whereas emergence from competitive, higher urgency contexts still exist, they just require a larger timeframe of technical exploration from the side of the player, in order to push the system to its limits. For example, another interviewee explains how their training mode sessions are solitary and exploratory: "Det var som at finde et eller andet slags sprog med sine karakterer. Af knapper og angreb. Og så bare kæde dem sammen i training mode alene var sjovt nok for mig. (see appendix, p. 24). This lack of perceived spectators allows for a different kind of immersive experimentation with emergence that is less performative, more intimate, possibly more innovative yet requiring more time. However, this also applies to non-competitive players, as one interviewee talked about beating up computer players in Street Fighter 4, and finding enjoyment in learning how to exploit their AI (see appendix, p. 24). Although they also enjoyed the experience of becoming better at the game, they also admit that it did not make them a stronger competitive player per se, as they would still easily lose to those practicing against other human beings.

> "Så det var meget bare at træne. Og på det tidspunkt var der ikke et formål udover at jeg havde den sådan glæde for mig selv at jeg kunne finde ud af det her. […] Jeg tror det var først når jeg kom ud til et sted som det her [FGCA]. Eller prøvede sådan aktivt at blive bedre. At man lige skal få det slag i egoet. (see appendix, p. 24).

However, they still talked about it with a sense of pride, as it was building an identity, they were proud of and is still able to use the casually acquired skill by performing in casual contexts for the sake of just having fun instead of winning (see appendix, pp. 24-26).

But again, this also depends on context, as a competitive player might discover a new strategy, that they wish to develop. Even if this curiosity turns out to be nothing more than a gimmick, it might just instead be something to use as a fun finisher, or something with high risk and low reward that you can use to develop your identity as a user. For example, given a context with an audience, an otherwise hardcore competitive player might risk doing an otherwise unsafe move that might cost them the game, if it leads to a hype moment that can build their identity and social standing. The presence (or absence) of an audience changes how the system is played, what is considered valid strategy, and how risk is perceived.

These dynamics matter for design. Emergent design must account for the expressive, social, and contextual factors that influence how and why players innovate. It is not enough to create possible spaces for new behavior; designers must also consider how users encounter those possibilities in public, semi-public, and private modes of engagement. Emergence is not only about discovering what the system permits, it is also about performing something that feels worth doing depending on the user's identity and context.

8.3. Temporality and Evolving Engagement

Temporality refers to the emergence of interactions over time. Time is an enabler, that allows users to experiment and play with elements or mechanics. Although emergent behavior may happen quickly, it has also been observed to accumulate gradually, through repetition, variation, and ongoing engagement. Games are temporal systems by nature, and players' relationships with them mostly exceed single sessions of play. Temporality shapes how emergence occurs, as strategies evolve, habits form, and new interpretations of mechanics or systems become apparent. Time allows for patterns of interaction to stabilize or shift, and for knowledge to be transmitted or transformed within communities.

This section argues that temporality is fundamental to the development and fostering of emergence as a medium through which new interactions occur, but also as a condition that makes such iterative development possible. Therefore, understanding emergence requires attention to how interactions and users' strategies persist and changes over time.

8.3.1. Melee and Ultimate

Among the different sub-genres present at FGCA, those that play platform fighters might be the ones that stick out the most. Although the core of the game might seem different from traditional fighting games, the main thing that sticks out is how the two most popular platform fighters are both from the *Super Smash Bros*. series. The most recent entry *Super Smash Bros. Ultimate* from 2018 and the older entry *Super Smash Bros. Melee* from 2001. 'Melee' is considered an older game, and despite there having been multiple entries in the Smash Bros. series between Melee and Ultimate; Melee still manages to be the most popular Smash Bros. game at FGCA for the moment being. When talking with Schadrac, he explained that the continuous relevancy of Melee is entirely due to community support, as the original intent of the developers was never to create a competitive fighting game, but rather a casual party game more akin to games like *Mario Party* (see appendix, p. 6). What started as a casual experience gradually evolved into one of the most technically demanding fighting games in the scene, as players noticed, learned, and pushed the game mechanics further than developers intended. Melee could be described as an unintentionally underdesigned game, considering its short thirteen month development (George, 2012). This meant that the game was full of gameplay bugs, many of which would go on to be adopted by the competitive playerbase. Now, many years later, the game's competitive life is still active and defined by these bugs; now reframed as essential techniques such as 'wave-dashing,' 'L-canceling,' 'Wobbling,' and so on. However, as mentioned by Schadrac, this evolution required continuous effort from the community to shape the game into what it is today. As Schadrac put it: "Hvis vi ændrer banerne, så bliver spillet mere til den måde, vi gerne vil spille på. [...] folk modificerer spillene" (see appendix, p. 18). With this, throughout decades at this point, the Melee community has been molding the game both sociologically through offline rulesets, but also practically through modding the game; re-coding it, to enforce these community developed rulesets automatically. In a competitive context, the players must strip the game of all its 'party game' elements to create a consistent, fair, and competitive environmental context. Schadrac told the researcher of this phenomenon as he explained how it was a 'classic' example of how the community regulates the game. The community considering it a 'classic example' shows that the phenomenon is temporally loaded and how emergent appropriations have cultural significance within the community; what emerged as responses to a want for competitive integrity developed over time to change the opinion and perception of what Melee is. This showcases that emergent outcomes shape the perception of its given context if given enough time to develop.

8.3.2. The Affordances of Temporal Appropriation

The case of Melee's community upkeep showcases how appropriations over time naturally facilitates an environment for meta-design. The system is not simply played; it is restructured and maintained by the community to reflect its values; providing an environment that makes the tacit observable. This communal process of maintaining Melee also exemplifies how temporality shifts the outcome of emergence. What instilled the Melee community to create and maintain custom rules for as long as they have, came from the observed enthusiasm within the scene; a wish to maintain competitive integrity in a game not otherwise designed for competition. This meant, that the game's large list of 'casual' game mechanics, such as a variety of stages, items and strategies were first controlled through agreements between players, but as the game grew older and the internet became more prevalent, modding of the game became the norm. This allowed players all over the world to use the same modded version of the game which now included the rules and adjustments that have been refined over many years of competitive play. The amount of setup required before a match highlight how these preparations functions not only as a technical task but as a ritual of legitimacy for those wishing to partake; it becomes a kind of gatekeeping, as those wishing to engage with the community must conform to the cultural standards already set in stone, before they can interact with the game in the given context. This seems to be the case, as a way of reinforcing a specific identity depending on the given context and manipulated by the community's values surrounding it. The values of the Melee players at FGCA have been developed over many years, based on the needs they have encountered due to the identities of those involved in the community context.

Ultimately, what this phenomenon within the Smash Bros. community has led to, is the birth of the game *Rivals of Aether 2*, which is a platform fighting game designed to be competitive first and foremost unlike the Smash Bros. series. According to players at FGCA, the game attempts to mix the competitiveness and technical complexity of Melee with the approachable aspects and quality of life improvements that the modern Smash Bros. Ultimate brings with it. Moreover, Rivals of Aether 2 is a game developed by members of the larger Fighting Game Community, meaning that they themselves have partaken in the context; making them aware of the tacit needs from modern Smash players, that have been yearning for something new that the Smash Bros. developers could not, or did not wish to deliver. This shows, that environments open for meta-design can make the tacit visible, as it gives developers an opportunity to unpack otherwise tacit user needs from within systems. Temporality lets users appropriate, which in turn showcases the tacit elements they otherwise would not be able to communicate. The strength here being, that it shows developers what sort of system improvements are in demand.

Rivals of Aether 2 released only a few months before the researcher's official visits, but they were able to observe the higher than usual attendees than other games. This was especially apparent when looking at the sign-ups for the monthly tournaments, where people compete for money prizes, maintaining higher numbers of competitors than other staple games of FGCA that have been around for years (see figure 9). The success of the game shows the effectiveness of allowing users to collaborate in this kind of context. And although Rivals of Aether 2 is an example of a full system having been developed, this does not mean that appropriations over time only tells us of the bigger picture, but it stands to argue that it also showcases developers what smaller scale interactions could be needed within a system in order to maintain user satisfaction. In this case, it was a matter of abstracting what made the mechanical bugs from melee beloved by the community and repurposing them in a modern and improved way, that respects the qualities that had emerged over time. All in all, this illustrates how a system's long-term viability is not necessarily tied to developer intent but to temporal user appropriation.

	oktober 2024	november 2024	januar 2025	februar 2025	marts 2025	april 2025
GGST	6	8	9	10	11	11
SF6	8	8	7	8	9	7
T8	16	15	11	16	15	15
SSBM	9	5	2	6	16	8
SSBU	8	8	7	6	8	4
ROA2	0	22	16	12	20	13
Total competitors	39	41	34	40	52	41
Total attendees	45	46	36	43	57	47

Figure 9 - FGCA Monthly Tournament Attendees (Provided by FGCA Board Member)

8.3.3. Motivator for Temporality

While temporality is argued to enable emergence, temporality still requires a motivator. Malte expressed when discussing his relation to FGCA as a community that "nu har jeg mennesker jeg sådan kan snakke med om mine progressioner. Hvordan jeg klarer det. […] Og så komme herhen og så blive tæsket 10 games i streg af en eller anden King spiller jeg aldrig har set før, er sådan 'naah okay', der er noget mere at lære her." (see appendix, p. 23). He expresses signs of motivation for why he chooses to continue to engage with the medium. As he puts it, there is more to learn, that he would otherwise be unaware of. During observations, another member of FGCA jokingly brought up the juxtaposition of becoming better at a fighting game, which leads to them being harder to enjoy with friends that otherwise do not aim to become competitively better at the game. A phenomenon that many other community members could relate to. It shows how motivation to continue exploring the system dissipates if the social context does not support it. If the context makes it less attractive to interact with a system, in this example due to the sociological imbalance in user skill, then motivation disappears and temporality becomes obsolete. The surrounding community context becomes a motivator that enables emergent outcomes to be shaped by temporality.

8.3.4. Mastery of a System

Although goals differ depending on individuals and identity, ultimately having a goal to work towards; a reason to perform, seems to be a big factor when trying to give purpose for exploring a system. It motivates users to explore a system seeking mastery; to appropriate, which ultimately spawns emergent outcomes. When talking about mastery in this context, it is not meant as full and total understanding and control of a system, rather the ability to fluently use certain mechanics or perform interactions without overthinking it. Mastery is being able to instinctively pilot parts of a system as if it was second nature to the user. However, building towards mastery requires time. For example, competitive players might stumble upon a curious interaction, but they will need time to develop it in order to ascertain whether they can develop a competitive strategy around it. You would often hear people casually ask their opponents questions during sessions of play at FGCA about what just happened on screen, or how or why a certain interaction between characters played out the way it did. These small emerging moments of un-expectancy is what leads to ideas of new strategies that will then need further exploration. The idea of spending time 'in the lab (laboratory)' was a phrase that came up in conversation almost every time the researcher spoke with people at the weekly gatherings. It refers to the time spent in a games training mode; a sandbox mode free of consequences, that gives the player access to tools that lets them dig into the game's system in order to further master certain aspects of the game. It is a mode that has become integral to playing fighting games to the point, where modern fighting games are practically disregarded, if their training modes are shallow or non-existent. Malte exemplifies this: "Jeg er typen der godt kan lide at optimise af helvedes til. […] Jeg kan godt lide at min-max. Og så udnytte systemet så meget som muligt" (see appendix, p. 29). Interestingly here, Malte describes the game as a system. The choice of word carries connotations and assumptions; it shows how identity shapes how digital artifacts are perceived. During the interviews, Malte specified his love for fighting games, "Men jeg elsker bare kampspil. Jeg elsker dem som systemer. Og som noget at bare fjolle rundt med. Elsker at gå ind i training mode og bare trykke på nogle knapper. Det synes jeg altid er hyggeligt. Lige meget, hvilket spil det er” (see appendix, p. 25). In this context, when referring to it as a ‘system’, Malte describes it as a set of building blocks of which they are free to manipulate. What Malte tries to achieve by manipulating these blocks, is a level of mastery that differs from what a competitive player such as Schadrac might aim for. This, coupled with conversations on-site, gives the idea that ‘labbing’ has become such a cultural cornerstone of the community, that a subset of players have emergent, that identify themselves as ‘lab-monsters;’ players who spend more time in the training mode rather than playing the actual game. Malte showcases their perceived identity as a lab-monster, based on the way they talks about the games. They show appreciation for a ‘system’, rather than talking about it like one usually would a game. When asked about why Malte likes spending time in training mode, they mention how the creative part of it makes it worth it. “Ofte, i hvert fald før jeg havde et community af en art, at dele det med. […] Så var det bare sådan det kreative i det” (see appendix, p. 24). The way they explain it, makes it sound like the time they spend exploring the game as a system, is to try and make sense of its mechanics like pieces in a puzzle; they look for ways to put these metaphorical building blocks together so they fit in a particular way they see appropriates. Although this is only layer one. Once the core elements such as how to jump, move and throw a punch has been mastered mechanically, that is when the creative aspect comes into play, enabling further layers of system exploration. Having a level of mastery acts as a lens through which the user understands the system and what is possible. Being able to both understand and then use this knowledge to develop emerging strategies can be referred to as ‘freestyling’, a term used by Schadrac to inadvertently describe this concept of mastery:

> “Ja, freestyling det er det samme som en person på et skateboard eller en løbehjul eller noget gør, hvor selvfølgelig kan du bare køre hurtigt på rulleskøjter, men du kan altid ekspresse dig selv ved at lære at bremse med dit side i stedet for den typiske måde.” (see appendix, p. 14)

In this context, Schadrac also highlights the importance of user expression, as mentioned earlier. The act of freestyling lets a user perform their identity through system interactions. Mastery is identity-building. Especially in emergent games, mastery means learning how to bend the system, not just follow it. If there were only one solution to the systematic building block-puzzles, that players like Malte are trying to continuously assemble, then freestyling would not be possible, as the system is then fully progressive; giving no leeway for emergence.

However, while fighting games may be deterministic when it comes to game mechanics, they are not progressive. They are complicated games, meaning that once general mastery has been achieved, then emerging cracks in the system will begin to reveal themselves given enough time. As Malte puts it: "jo ældre og jo mere broken og *kusoge* det er, jo bedre bliver det" (see appendix, p. 23). Here, Malte shows appreciation for a sub-genre of fighting games called *kusoge* (The Fighting Game Glossary, n.d.), which literally translates to 'shitty game' from Japanese. It is a somewhat endearing term used by the fighting game community used to describe highly broken fighting games that end up being fun to play, despite the underdeveloped design or numerous bugs. However, players with a sufficient level of mastery like Malte, can appreciate this type of game, as they are aware of how the broken parts of it affects potential emerging strategies, and can thereby more easily freestyle strategies to push the games to their limits. In this sense, level of mastery is dependent on temporality as a factor. This shows that time spent exploring and learning a system is not just to hone mechanical skills, instead these interactions shaped over time determine the user's ability to freestyle interactions within a system of rules. By letting users master a system, the more they are able to freestyle, which then leads to emergent outcomes.

9. Findings Continued: How Context, Identity & Temporality Shape Emergent Outcomes as Situated Conditions

Based on the presented empirical findings from the field study, it has been identified that context, identity, and temporality are key mutators that define and shape emergent outcomes during user interactions with a digital artifact. They were earlier referred to as carriers of tacit knowledge, which we can now look at from the perspective of them not just being elements that describe what the players do, but instead how they embody the player's understanding and how they navigate the interactive challenges they wish to overcome through a digital artifact. Therefore, this paper will be categorizing and labeling these three components as *situated conditions* inspired by Suchman's (1987) idea of situatedness in interactions. This is to reinforce the idea that emergence is not as abstract as originally perceived, but instead an embodied phenomenon that evolves based on user interactions and influenced by these situated conditions as variables, made discoverable as users' tacit knowledge or behavior. It is not unlikely that there are more situated conditions than what was discovered during the field study of this paper, however for the time being it seems realistic that *context, identity,* and *temporality* act as the main three actuators in the equation, and they will therefore be looked at first and foremost.

9.1. Context – Immediate & Extended

As established, emergent outcomes do not happen in a vacuum; they are situated phenomena that are defined by the material and social conditions of the structural context at play. Calling it structured context is to emphasize that these contexts are not fully in the users' control as although context changes over time, it is also pre-existing from the user's first point of entry. From this, we can understand context in two parts that this paper terms: *immediate context* and *extended context*; both structured by the tacit norms known to those individuals who are an active part of the context.

The immediate context consists of the environment; the material and spatial conditions for interactions. This includes how the physical location, available devices and social presence determines the context. For example, as previously observed, offline play at FGCA compared to online play offers a more potent immediate context for emergent outcomes as it allows for the use of body language and informal conversations that does not flow the same way online. While some interactions are available both offline and online, the context changes how they are both performed and perceived, which results in varying user agency and decision making.

Then, the extended context refers to the larger social and cultural structures that define the context. This includes the greater cultural contexts if the context is a sub-culture of something else. FGCA is a sub-community within the grander Fighting Game Community, which means that they are also influenced by other sub-communities, which in turn also affects the Fighting Game Community as a whole. Tacit community norms such as what is considered “legit,” “toxic,” “hype,” or “meta” differs from each sub-community, but is also influenced as a whole. The extended context is also situated in its cultural meaning, for example, what does Super Smash Bros. represent as a game series? For outsiders, it might be considered simply a party game, but others might see it as a fighting game. These broader elements shape the interpretive frame through which interactions are understood. For example, associating and calling oneself a ‘fighting game player’ carries assumptions that complicate the user’s context and thereby shapes the interactions and emergent outcomes.

9.2. Identity – Identity Performance

Identity within emergent systems can be understood through what this paper terms *identity performance*. This concept captures how users enact and express identity through their interactions with digital systems. These performances are not fixed but fluid, shaped by user intentions, social context, and evolving engagement with the system. Users construct their identity in relation to the system they interact with. Both in how they interact with the system, but also in regard to how they position themselves in relation to the system. A user’s identity not only affects how they engage with a system, but also affects how those within the same context interact based on the expressions of other individuals. Identity also brings motivation and shape to emergence. Users do not engage with systems as abstract agents, instead, they bring goals, values, personas, and social affiliations with them. The field research showed how the players at FGCA used their moments of interaction to express their personalities, which in turn changes how they approach their system interactions and user journey. This goes to show how identities in the context of interacting with a digital artifact are performed and not necessarily possessed, in the sense that they are not static, but instead developed through context and social framing. They might showcase signs of being an innovator or explorer, a competitor, a casual, a community enthusiast, or something completely different. The user's identity gives them purpose in their interactions and in turn, they perform their identity through use of the digital artifact. Therefore, identity performance shapes how emergent outcomes are molded. Emergent outcomes often reflect this in return: players make the system theirs, as a form of identity performance and expression. These identities inform what users notice, what they value and how they choose to appropriate affordances. This information is tacit yet crucial for understanding user needs, and emergence then becomes a lens which abstracts the information into something observable.

9.3. Temporality – Modes of Temporal Emergence

On the surface, temporality is observed as mainly a modifier of other conditions, however it also enables emergence by letting patterns develop through repeated interactions. It allows for experimentation and adaptation. Emergent outcomes begin as exceptions or improvisations; as reactions to moments of novelty, and then gain structure and legitimacy over time and repetition. As observed, game communities such as FGCA develop rules and norms this way; emergent outcomes because of accumulated user behavior throughout time. The more uses or the more time that passes, the higher the chance a user might experience something unexpected. A player at FGCA's weekly gatherings explained, that they refrained from using over-tuned strategies or moves, as they were certain they would later be removed or changed. This was to avoid letting themselves develop bad habits for later iterations. This is an example of a user's emergent strategy being developed on tacit knowledge. When discussing temporality in the context of emergence, we can analyze these types of thought processes as user identities shaped by *modes of temporal emergence*. What is meant by modes of temporal emergence, is that emergence was observed to happen at varying temporal scales. Firstly, we saw that emergence can happen as *in-the-moment emergence*, mostly in non-consequential or casual contexts among the users' peers. These are moments of emergence, that are taken at face-value; they disappear as quick as they emerge, often tacit, and are therefore harder to notice unless you are an avid user of the system the interaction emerged within. Second, we observed *continuous emergence*; instances of novelty that trigger curiosity, which will need time to develop. Continuous emergence is defined by its iterative qualities, as they are developed through continuous use and exploration. They are moments of emergence that appear multiple times but every time a little different, as they are subconsciously iterated upon and explored in future interactions based on user identity and context. Alternatively, we also saw *determined emergence*. We observed how competitive players notice in-the-moment emergence, and actively decide to explore it further. While reminiscent of continuous emergence, this type requires pre-exploration. For example in the form of training an emerging skill before a big match; before they must perform. What sets it apart is the user's awareness of knowing that they must eventually perform, whereas continuous emergence is less reliant on potential consequences as a motivator, hence why they choose to spend so much time actively pushing in-the-moment discoveries to further develop their identity in the context of the system. Each mode of temporal emergence has its benefits:

- In-the-moment emergence helps reveal tacit system affordances while informing about peripheral user behaviors that could become more meaningful through further support.
- Continuous emergence illustrates the subconscious interactions with a system that the user wishes to engage with. The emergent outputs that the user instinctively continues to attempt to make work, should be considered by designers when iterating the system.
- Determined emergence helps by showcasing the user's motivation by framing their journey through the system; their pathway towards mastery. It gives insights into users' tacit knowledge by showing what the user actively wishes to engage with within the system, which helps designers understand what parts of a system affords what levels of agency from the user.

With this in mind, different modes of temporal emergence become a constitutive factor that allows for emergence to develop from something unintentional and evolve into something intentional. It lets the artifact showcase interactive possibilities through user expression. Emergence happens over time, not in a single moment. Users evolve, systems evolve, contexts evolve, and Emergent design acknowledges the fluid, time-sensitive nature of user engagement. Designs able to acknowledge this supports that trajectory and creates better environments for fostering innovation.

9.4. Situated Conditions as Influencing Variables

During user interactions, the user provides an input that will eventually be returned as an output. Outputs can be in line with the expected progression; however, most outputs are inevitably going to be unexpected and thereby emergent outputs. When an interaction does produce emergent outcomes, it is because the output has been sufficiently shaped by the situated conditions. But then, in theory, the emergent outcomes should stagnate as the conditions stay the same. But, although situated conditions shape emergence, they are also changing variables and are affected by each other. While presenting the ethnographic findings, an attempt was made at trying to keep the three situated conditions separated while discussing them, although this proved impossible as it quickly became clear that all three conditions affect each other directly. For example, as mentioned earlier, even if context is structural, it is affected by temporality and changes over time. At FGCA, the members change; newcomers join and other members might go on hiatus or leave permanently. The venue might change to a different building with different conditions, or it might drastically switch to online only. Essentially, temporality affects context. With this in mind, a pattern emerged and its effects became apparent. Therefore, we can present an illustration (see figure 10) of this phenomenon and how each condition might affect one another:

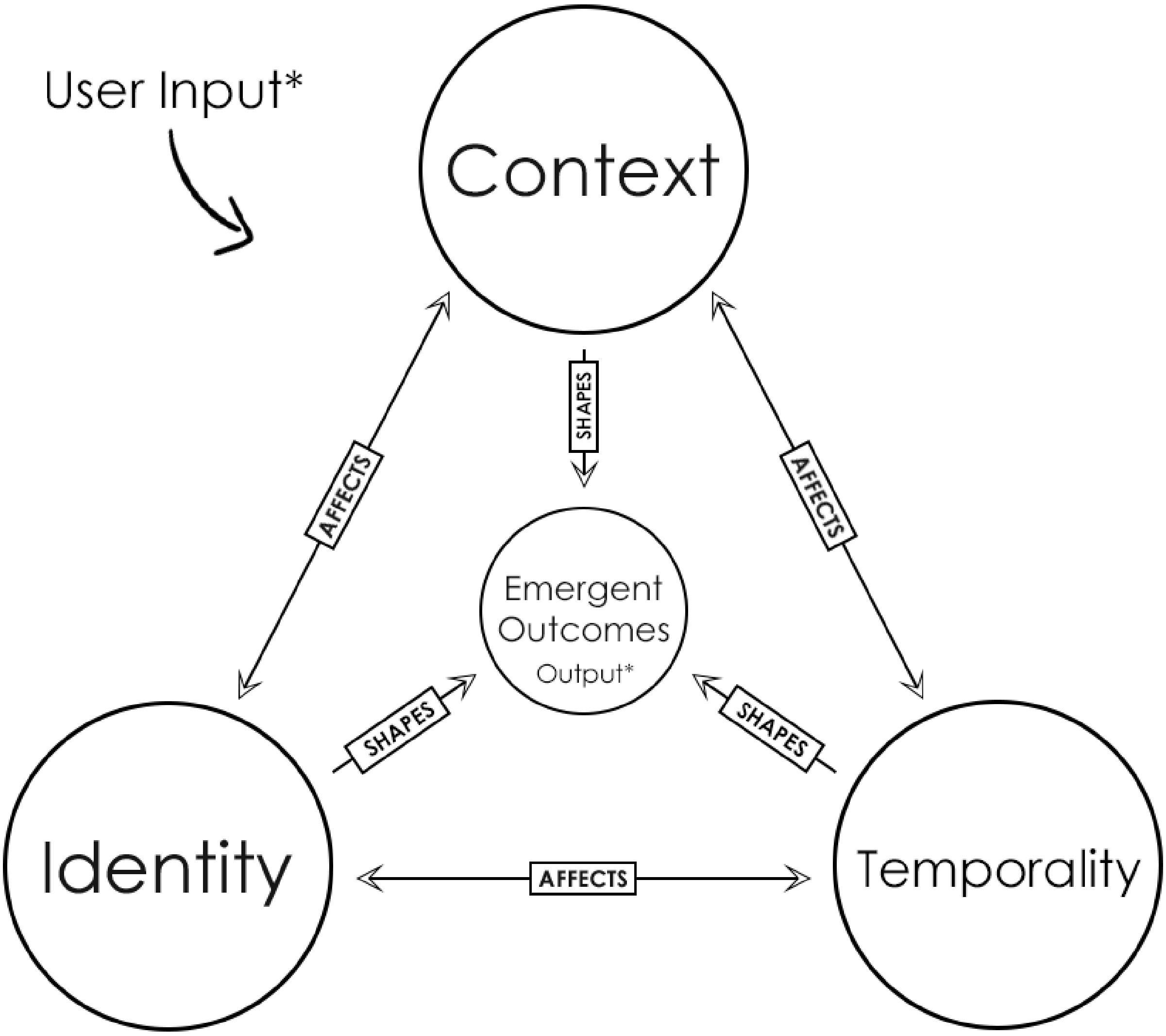

Figure 10 - Illustration of Situated Conditions

1. ***Temporality* affects *Context*** by reshaping how context is structured and perceived over time. Design plans may set initial expectations, like expected emergence and progression, however sustained usage of a digital artifact reframes the social and physical context of interactions.

2. ***Temporality* affects *Identity*** by allowing for identity to change. The more a user engages with a system, the more it changes their perception and how they further interact with it. Over time, a user might go from a casual user to an avid inventor or similar. The user’s role changes as their interactions evolve.

3. ***Context* affects *Temporality*** by shifting the affordances and constraints based on the context (e.g., device availability, social norms, local rules), which determines what kinds of temporal practices are even possible.

4. ***Context* affects *Identity*** by structuring the roles available to users and shaping how those roles are valued. Contexts are not neutral as they carry social expectations and hierarchies that inform how users see themselves, perform and how they are perceived. The identity of a 'Smash player,' a competitor, or a tournament organizer is not simply personal preference; it emerges in part from how those roles are legitimized within the situated environment. Context both affords and constrains the performance of identity.

5. ***Identity* affects *Temporality*** by orienting users towards particular rhythms and goals. A user might identify as a competitive player which in turn affords longer play sessions through daily grinding as they search for optimization. In this sense, identity is a temporal lens, that filters how users construct the duration and progression they prefer from the system.

6. ***Identity* affects *Context*** by shaping the context to fit the variety of users that interact within the relevant circumstance. For instance, the transformation of Smash from a party game to a competitive game is not inherent to the original system or developer intention, but the result of players asserting different identities within it.

Finally, these things culminate in the symbiotic triangle of situated conditions that affect each other and finally shape the emergent outcome of user interactions (see figure 10). Rather than treating emergence as a fixed system, this perspective understands it as a product of how users inhabit and interpret systems within specific environments, over time, and through ongoing identity performances.

10. Reviewing the Initial Framework

Before moving on to presenting the second framework iteration, we can look back at the first iteration and compare it to the empirical findings to see what aspects might still be usable moving forward:

The first framework presented four design dimensions: the environmental context, the level of expected emergence, user agency and consequences and designing for underdesign.

Environmental Context

The first iteration of the emergent design framework successfully identified the concept of environmental context. It correctly anticipated that emergent behaviors are more likely in co-located settings. However, as seen through the empirical findings, this dimension lacked sensitivity to the social norms and validation mechanisms embedded within those environments. As with Schadrac's shy attack cancelling, the move only became valuable after it was recognized by peers; not as a meta-defining technique, but rather as a fun, identity-defining choice.

The Level of Expected Emergence

While this concept initially showed potential through theoretically relevancy, it remained somewhat underdefined. However, the field research provided us with the idea of modes of temporal emergence, where expected emergence can be used as a modifier for which mode is relevant. As originally outlined; expected emergence was introduced as a designer's tool for calibration of how much unpredictability a system is intended to allow. Then, seen from discussing temporality as a situated condition, modes of temporal emergence instead describe the actualized unfolding of emergent outcomes across users' interactions over time. Expected emergence must be understood not as a fixed property, but as a dynamic variable that plays out through modes of temporal emergence. The designer's task is not only to set levels of expected emergence but to anticipate and monitor its realization as the tacit becomes apparent or evolves through usage cycles. These two ideas can effectively be linked in order to let designers anticipate what type of temporal emergence certain affordances might produce.

User Agency and Consequences

The initial ideas about agency and consequences as motivators for user interaction remain relevant, however the ethnographic findings reframe it from functional consequences to social consequences. Context shapes motivations for user expression. This came from observing examples of identity performance, such as Schadrac’s home-developed technique, which emerged not from the exploration of system mechanics alone, but also from how it was perceived by others. The concept of "Performing Perception" helps highlight how the social pressure to perform and have their identity socially verified influences players’ emergent behaviours.

Designing for Underdesign

The approach of underdesigning continues to seem promising. Although the field research was unable to directly test the method, observed examples such as the success of Rivals of Aether 2 and the appreciation for the “kusoge” sub-genre, shows that when gaps in design are embraced, they can create spaces for experimentation and creativity that breed engagement for emergent outcomes through the freedom of appropriating a system. This ultimately leads to systems with increased user satisfaction, such as Rivals of Aether 2.

11. Second Reworked Framework

This framework, like the first, will be created through the intent of giving designers a methodological and iterative approach to designing and creating digital artifacts through human-centered design values. The iterative aspect is important due to the nature of emergent outcomes. Therefore, this new framework will present a repeatable process divided into stages rather than general guideline like how the initial framework did it. The framework is also built with the intention of being repeated. It should be used to gain insights that will reframe itself in future iterations of the same system. The intention is for every iterative loop to give new emergent outcomes from which the designer can find qualities to develop for the system in development.

The framework assumes that the design process is far enough to allow for iterative prototyping. The technical fidelity of the prototype is irrelevant, as long as there is a layer of interaction available for users to engage with to allow for co-designing. Again, it is based on the theoretical exploration originally made, but reframes the concepts through the ethnographic findings which it also prioritizes.

11.1. Framework Overview

<u>The stages of the framework will be as follows:</u>

Stage 1 - *Explore context and user group* →

Stage 2 – *Prototype adjustments* →

Stage 3 – *Activate user engagement* →

Stage 4 – *Gather data* →

Stage 5 – *Analysis and reframing*.

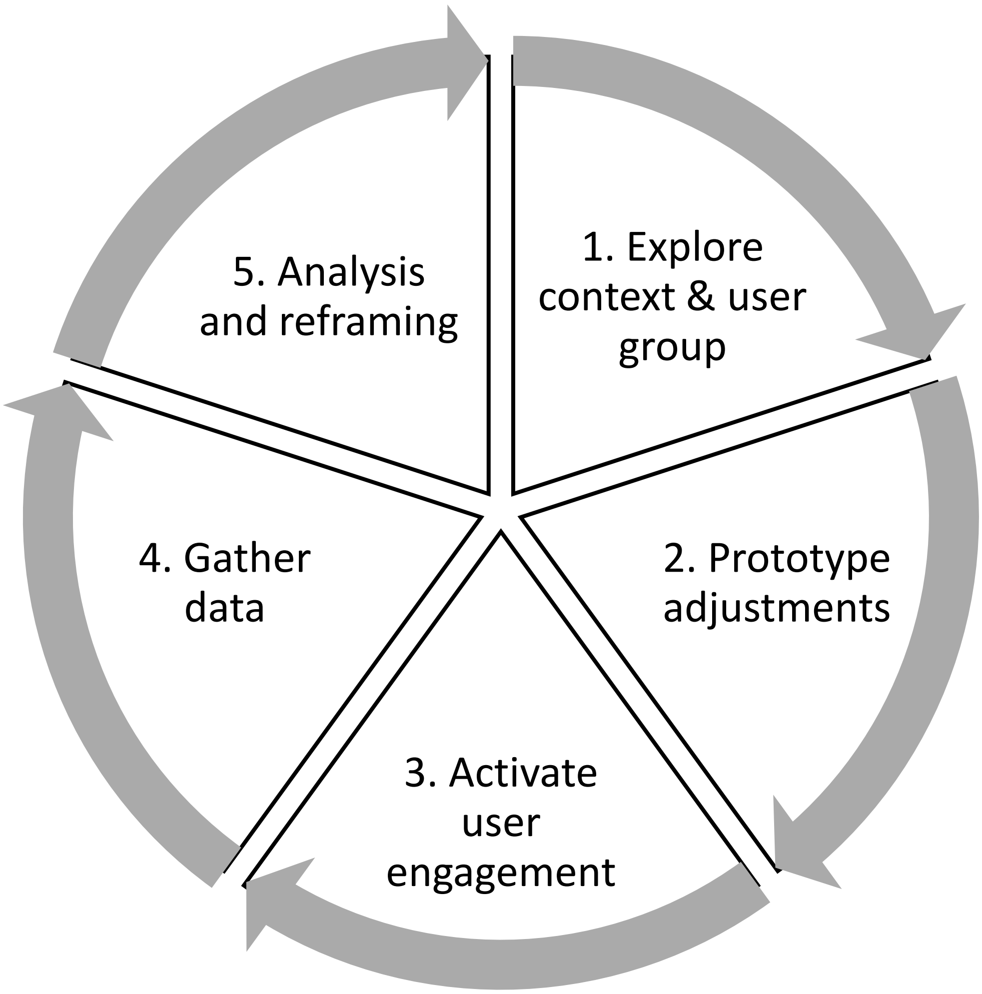

Figure 11 - Illustration of Second Reworked Framework

11.2. Detailed Framework

Stage 1 - Explore context and user group.

The first stage centers on contextual and user understanding to define the designer's expectations for what is to come. Begin by situating the design in its real-world context and community, recognizing that emergent behavior is shaped by the environment and the people involved. The designers conduct ethnographic research to learn about the context (who are the user group, their activities, and the social setting and physical environment). Designers then empathetically engage with users within the context (through observations or interviews, etc.) to grasp these situated factors, consistent with human-centered design's emphasis on real-world understanding and empathy. Crucially, the team also considers temporality; the effects of use over time. Emergent behaviors often gain structure and legitimacy over time and repetition, so the process is planned to allow multiple usage cycles rather than a one-off deployment. By the end of stage 1, the designers should have a grounded understanding of the context and users, have framed initial design goals, including how much emergence they intend to encourage.

Strategies:

- Study both the immediate and extended context (space, culture, community, etc.) through ethnographic research to identify how context might shape emergent outcomes.
- Understand users and their identity by engaging with them through ethnographic research to learn about their identities and thereby motivations and current skills, in order to inform how they interpret system affordances.
- Based on the insights gained from the first stage, define the initial emergent design strategy, by deciding the expected level of emergence in accordance with the expected modes of temporal emergence. Designers should not only ask "*how emergent should this be?*" but also "*when and through what modes of temporal emergence do I expect emergence to unfold?*"
- Plan for recursion. This is an iterative process, so expect to be back at stage 1 again.

Stage 2 – Prototpye adjustments:

This stage involves creating or iterating on the current prototype. The guiding principle here follows the idea of designing for underdesign, by intentionally leaving parts of the prototype open to interpretation through incomplete or loosely defined affordances which should nudge the user towards appropriation of the underdesigned affordances.

Strategies:

- Deliberately leave certain features open to user modification or interpretation, rather than over-engineering every detail. The less the better initially, then tighten the different interactive elements as the prototype develops.
- Empower users to play and experiment with the prototype. Users repurposing parts of a system is telling. By designing an incomplete or modular system, the prototype can adapt to user creativity. For example, provide tools or settings that players can tweak, or optional rules that can be turned on/off.
- Readjust the level of expected emergence. Articulate the hypothetical limits that you wish to aim for. This includes the limits of the system and what aspects you expect to be flexibility. For instance, if high emergence is desired, the prototype might use less or minimal linear guidance, whereas in parts requiring less emergence, more structure is provided. Designers should treat this as a hypothesis, knowing it can be adjusted later.

By the end of stage 2, the prototype should be developed into an interactable artifact with open affordances that match the planned expected emergence to a point, where it is ready to be introduced (or re-introduced) into the users' hands.

Stage 3 – Activate user engagement:

This stage puts the prototype into the hands of users in a real usage context, enforcing the values of HCD in the design process by turning it into a co-design activity. Here, users' interactions and adaptations contribute to the design of the system. Users are partners in development. Preferably, the designers should let users interact with the prototype in the relevant context (or invites users into a realistic setting) and encourages them to engage with it naturally, as they would in real life.

Strategies:

- Let the users take initiative. Interject only when absolutely necessary. Users' frustrations are signs of where emergent strategies might be required of them.
- Encourage users to "break" the system. Invite users to treat the system playfully and modify or explore it as they see fit. Make it clear that unconventional use is welcome. In this way, users' improvisations become design inputs.
- Ensure users are not distracted with what they "should" be doing. We are instead interested in what they "can" do.

Stage 4 – Gather data:

This stage runs concurrently with stage 3. It is dedicated to data gathering. Here, the designers adopt an ethnographic stance in order to document how users interact with the prototype and what emergent outcomes appear.

Strategies:

- Consider what types of temporal emergence you expect and adjust data-collection strategies based on the assumption. Are you looking for in-the-moment emergence? Then actively collecting feedback and ideas while users engage might be beneficial, whereas exploring determined emergence might require multiple post-engagement interviews over a period of time.
- Log emergent behaviors. Some types of emergence happens quicker than others, so ensure you can log at least *what* happened and under what conditions (context, who was involved, when did it occur, etc.)

- Record how users react to outcomes. Are they engaged, frustrated, confused, competitive, motivated, etc.? Positive or negative emotions accompanying engagement gives insight into what value the outcome might provide.
- Pay special attention to those parts of the system that were intentionally left underdeveloped.
- Observe how the environment and social context played a role. Did being in a group change behavior? Did physical layout or devices used limit or enable certain interactions? These insights ensure the next design iterations are better situated the actual use context.

Stage 5 – Analysis and reframing:

In the final stage of the cycle, the designers should analyze the collected data and abstract them to tweak the next design iteration. This stage is about making sense of the emergent outcomes and using them to refine the design. The goal is to identify knowledge that can be made actionable: what did we learn about user behavior, needs, or preferences that was not apparent before? What should be supported in the design moving forward? Conversely, which issues or negative outcomes should be addressed? At this point, the designers revisit the original assumptions from stage 1 and the decisions made in stage 2. With empirical evidence in hand, some assumptions will be validated, while others may need reframing. Using the analysis, the team then devises concrete design changes for the next iteration. Desirable emergent behaviors identified during this stage might be explicitly supported in the new design. Undesirable behaviors are mitigated by adjusting constraints or providing better feedback. The designers also update the expected emergence. Essentially, the learnings are fed back into the design: every emergent outcome is treated as user feedback for improvement. The situated conditions are considered here as well: context, identity, and temporality insights guide which changes will improve user experience. This step is essentially design evaluation through an emergent lens, turning the experiment of stage 3 into concrete decisions.

Finally, with all this in mind, the designers iterate by returning to stage 1 (or stage 2, if context has not changed. Although chances are it will as things constantly change) in order to produce the next iteration of the system then repeat.

Strategies:

- Be determined. Even if the first tests yield less usable outcomes, it may just be, that the system affords modes of emergence that requires more time to develop, before the outcomes become apparent.
- Filter and decide by prioritizing which emergent outcomes to integrate or discourage. If an emergent strategy enhanced user satisfaction, then plan to formally support it in later iterations for further testing.
- Modify design elements and update requirements based on the above decisions as the reasons for design elements are now informed by empirical insights.

12. The Value of Emergent Design as a Methodological Framework

The findings of this paper, and by extension the framework, indicate that emergence is not just a phenomenon to be observed, but a considerable approach for how systems can be designed. In this case, by focusing on fighting games as the digital artifact in question, it became evident that user behavior has the high chance of extending beyond what is planned by designers. Players adapt, reinterpret, and restructure the systems they engage with, often revealing opportunities the original design did not account for. Emergence, then, becomes a source of design knowledge.

With the benefits showed, the practicality of such a framework in commercial design settings becomes the question. Especially in UX work with tight deadlines and fixed deliverables, designing for emergence may seem like a luxury. Iteration cycles are shorter, and stakeholders often prefer the safety in predictability. Yet this is also where the concept is most needed. Users regularly encounter systems that are overdesigned, heavy with features that try to cover every possible use case, but miss the flexibility to adapt to what users actually want. In practice, embracing underdesign through emergent design means resisting the impulse to account for every contingency upfront. Instead, designers provide a flexible framework or toolset and trust users to co-author some of the solutions, like how the Melee community did by developing mods and house rules to refine their gaming experience. An approach grounded in underdesign, as presented, can create space for more adaptive, situated outcomes. Crucially, underdesign does not imply a lack of care or quality; it is a purposeful incompleteness that invites user participation. Fischer and Giaccardi’s (2006) meta-design principles support this idea: since not all future scenarios can be anticipated, systems should be built to evolve through use. In commercial UX, eve if a company cannot fully crowdsource its design, it can still learn from user adaptations and incorporate the most valuable emergent outcomes in updates. In this way, emergent design principles can be incrementally applied to make user-centered iterations an ongoing part of the life cycle of a digital artifact, even under business constraints.

It is also worth situating the philosophy of underdesign in emergent design within larger contemporary movements in design and technology. The push against overdesign reflects a growing sensitivity in HCI and design communities to issues of sustainability and what Selwyn (2023) describes as "digital degrowth." Digital degrowth advocates for scaling back the relentless addition of features and reducing technological bloat in favor of more sustainable, human-scaled solutions (Selwyn, 2023). An emergent design mindset supports degrowth values through the development of systems that are more attuned to users' practices, but also more sustainable, because they evolve in response to real use rather than speculative requirements.

The framework presented in this paper encourages a balance through the lens of expected emergence: structure where it is needed, and openness where user input might offer value. While this thesis focused on video games, the core ideas apply to broader systems. Emergence still happens in these spaces. Recognizing it, and making room for it, may lead to systems that are not only more responsive but also more sustainable in the long term. Rather than treating the unexpected as a problem, emergent design treats it as a resource. That shift in perspective is where its real value lies.

13. Future research

A lot of time has been spent arguing for the relevancy of an emergent design framework and how one might function. However, the main caveat with it this far into the process, is simply that it has not been tested in action. It is built on theoretical hypothesis and data abstractions. Given more time, the next step in the process would be to go use it as the basis for developing a design case. A new framework like what has been presented needs iteration itself, which should be clear in how it changed from the first to the second iteration. It being in early development also means, that it has potential to grow in unexpected ways, which of course should be embraced, given the matter at hand.

Furthermore, the empirical work conducted has provided insights into how emergent outcomes are shaped. Still, these insights are grounded in a niche context: competitive fighting games. Although many of the findings are likely to be applicable to other digital systems and artifacts, further studies are needed to evaluate whether the framework maintains relevance in other system contexts. Based on this study's findings, they seem applicable, but this remains speculative without further empirical support.

Moreover, the framework promises an ability to help designers anticipate, frame, and utilize the unpredictability of emergence as a quality. This could be useful for peers working in fields where user behavior regularly exceeds design expectations. By offering a structure for working with emergence, the framework could support more resilient and user-aligned design practices. But this utility must be validated through applied research. Future studies should explore whether this methodology improves user engagement, satisfaction, or system adaptability when used in active development contexts.

Therefore, I implore other researchers who might have been persuaded to believe the relevancy of emergent design to attempt to use the framework or at least include the apparent value that lies in emergence within their own projects. So will I, in future work, as emergence is unavoidable. Even in the most rigid, deterministic systems, emergence will still happen, and emergent design could be the approach needed to frame and manage the unexpected for better user experiences.

14. Conclusion

This paper has examined emergent design within interaction design, repositioning it as a structured methodological approach. The research began by establishing a theoretical foundation and clarifying the definition of emergent design in the context of interactive digital systems and video games. Building on this groundwork, a first-iteration emergent design framework was developed to guide later analysis. An ethnographic field study within Fighting Game Community Aarhus was conducted, where user-driven emergent outcomes were observed and documented through ethnographic observations and group interviews. By analyzing these emergent behaviors within their contexts, the study gained empirical insight into the dynamics of emergence and finding three main situated conditions (context, identity and temporality) it considers to be essential in provoking useful emergent outcomes. These findings informed a refinement of the initial framework, aligning it more closely with the ethnographic data. Through this iterative process, this paper demonstrates that emergent design can be intentionally harnessed: what often begins as an unintended outcome can be recognized and incorporated as a deliberate design strategy. The key contributions include an empirically grounded framework for applying emergent design principles, and a showcase of the value that emergent outcomes add to user experience. By repositioning emergent design as a viable, structured approach rather than a byproduct, the study addresses a notable gap in interaction design theory and offers a nuanced perspective on designing for iteration and complex, evolving user interactions.

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